

PRESIDENCY UNIVERSITY BENGALURU

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

**CSE2252 DATA COMMUNICATION AND COMPUTER NETWORK
LAB MANUAL**

III B.Tech 3rd Semester A.Y (2025-26)

Instructor Incharge : Ms. Alina Raheen, Ms. Sterlin Minish T N, Dr. D. R. Denslin Brabin

Course Credit Structure : 0-0-2(1 Credits)

Experiment 1

1(a) Study of basic network commands and network configuration commands.

1(b) Identify and explore Network devices, models and cables

1.PING Command

How to check internet connection in CMD

To check whether your internet connection works, you can use Command Prompt to test your connection to a certain website or internet location. To do that, you can use the ping network command, followed by a web address or IP address. For instance, you can check the connectivity to GOOGLE without opening a web browser, by typing the command " ping www.google.com." Then press Enter on your keyboard.

Ping is used to check the connectivity with other devices on the network, for example computers, routers, switches etc. Select Start > Programs > Accessories > Command Prompt. This will give you a window like the one below.

Type *C:\>ping x.x.x.x*

By default, ping sends four ICMP Echo Request packets each of 32 bytes. The response packets are called ICMP Echo Reply Packets.

A screenshot of a Windows Command Prompt window. The title bar reads "C:\WINDOWS\system32\cmd.exe". The window content shows the following text:

```
Microsoft Windows [Version 5.2.3790]
(C) Copyright 1985-2003 Microsoft Corp.

C:\Documents and Settings\Administrator>ping 155.0.0.24

Pinging 155.0.0.24 with 32 bytes of data:

Reply from 155.0.0.24: bytes=32 time<1ms TTL=128
Reply from 155.0.0.24: bytes=32 time<1ms TTL=128
Reply from 155.0.0.24: bytes=32 time<1ms TTL=128
Reply from 155.0.0.24: bytes=32 time<1ms TTL=128

Ping statistics for 155.0.0.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure1: The Ping Command

Now Type *C:\>ping x.x.x.x -t*

- The ping command also allows you to use the handy "-t" parameter, which enables you to ping the specified address forever until it's manually stopped. For instance, we typed "ping -t www.digitalcitizen.life." After some time, we wanted to see some connection statistics and we used the keyboard combination "CTRL + Break." This shows the averages of the ping commands run until then.
- "-t" switch will continue to send packets to the destination until user stops this by pressing *Ctrl + C*

```
C:\Users\Codrut Neagu>ping www.digitalcitizen.life -t

Pinging www.digitalcitizen.life [2606:4700:20::681a:cbc] with 32 bytes of data:
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=85ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=18ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=11ms
Reply from 2606:4700:20::681a:cbc: time=21ms
Reply from 2606:4700:20::681a:cbc: time=64ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=14ms

Ping statistics for 2606:4700:20::681a:cbc:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 85ms, Average = 25ms
Control-Break
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=10ms
Reply from 2606:4700:20::681a:cbc: time=10ms

Ping statistics for 2606:4700:20::681a:cbc:
    Packets: Sent = 16, Received = 16, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 85ms, Average = 19ms
Control-C
^C
C:\Users\Codrut Neagu>
```

2. IPCONFIG Command

How can I see all the network adapters on my computer using CMD?

To obtain detailed information about your network adapters and connections, use the ipconfig command. Open Command Prompt, type ipconfig, and press Enter. As you can see in the screenshot below, when you run this command, Windows displays the list of all the active network devices, whether they're connected or disconnected, and their IP addresses. You also get details such as their default gateway IP addresses, subnet masks and the state of each network adapter.

```
Command Prompt
C:\Users\Codrut Neagu>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2a02:2f01:730a:1300:107c:de5c:5f89:c00a
    Temporary IPv6 Address. . . . . : 2a02:2f01:730a:1300:254b:7d03:4a72:9b5c
    Link-local IPv6 Address . . . . . : fe80::107c:de5c:5f89:c00a%20
    IPv4 Address. . . . . : 192.168.50.239
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6d9:f5ff:feb5:b1f0%20
                                192.168.50.1

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\Codrut Neagu>
```

Displays full TCP/IP configuration of all network adapters (Ethernet cards) installed in your system. Type the following command in the command prompt.

C:\ipconfig

```
C:\WINDOWS\System32\cmd.exe
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.5.28
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.5.100
```

Figure 2: The IPCONFIG Command

Now type *C:\ipconfig /all*

If you add the /all switch to the ipconfig command, you can get to a whole new level of detail: DNS information, the MAC (Media Access Control) (in the Physical Address field), and other information about each network component. Check out the picture below to see a sample of what you get from the "ipconfig /all" command.

```
Command Prompt
C:\Users\Codrut Neagu>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Codrut-PC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe 2.5GbE Family Controller
Physical Address. . . . . : 04-D9-F5-34-B1-A3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2a02:2f01:730a:1300:107c:de5c:5f89:c00a(Preferred)
Temporary IPv6 Address. . . . . : 2a02:2f01:730a:1300:254b:7d03:4a72:9b5c(Preferred)
Link-local IPv6 Address . . . . . : fe80::107c:de5c:5f89:c00a%20(Preferred)
IPv4 Address. . . . . : 192.168.50.239(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, January 23, 2020 1:45:35 PM
Lease Expires . . . . . : Friday, January 24, 2020 1:45:34 PM
Default Gateway . . . . . : fe80::6d9:f5ff:feb5:b1f0%20
                             192.168.50.1
DHCP Server . . . . . : 192.168.50.1
DHCPv6 IAID . . . . . : 335862261
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-21-90-1C-04-D9-F5-34-B1-A3
DNS Servers . . . . . : 2a02:2f01:730a:1300::1
                             192.168.50.1
                             2a02:2f01:730a:1300::1
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 38-00-25-41-C3-F5
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Ip config has a number of switches the most common are:

ipconfig /all - displays more information about the network setup on your systems including the MAC address.

ipconfig /release - release the current IP address

ipconfig /renew - renew IP address

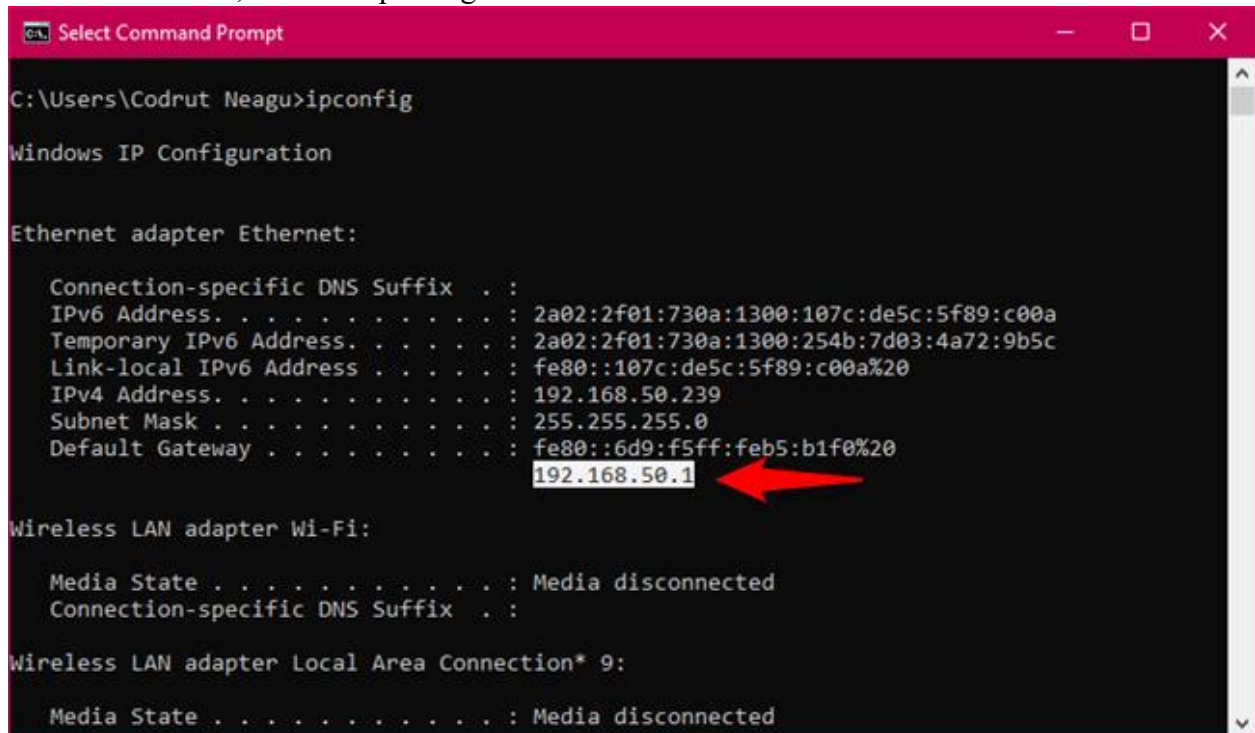
ipconfig /? -shows help

ipconfig/flushdns - flush the dns cache

How to check your network connection in CMD

If you want to check whether your network connection to the router is operating as it should, you can use a combination of the commands **ipconfig** and **ping**. First, get some cmd nic info about your adapter. In other words, open Command Prompt and run **ipconfig**. In the list of results, identify the

network adapter that's used for connecting to the network you want to test. Then, in its details, find the IP address of your router and note it down. For example, if we'd want to check our Ethernet network connection, we'd run ipconfig and see that our router's IP address is 192.168.50.1.



```

C:\Users\Codrut Neagu>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2a02:2f01:730a:1300:107c:de5c:5f89:c00a
    Temporary IPv6 Address. . . . . : 2a02:2f01:730a:1300:254b:7d03:4a72:9b5c
    Link-local IPv6 Address . . . . . : fe80::107c:de5c:5f89:c00a%20
    IPv4 Address. . . . . : 192.168.50.239
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6d9:f5ff:feb5:b1f0%20
                                192.168.50.1

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

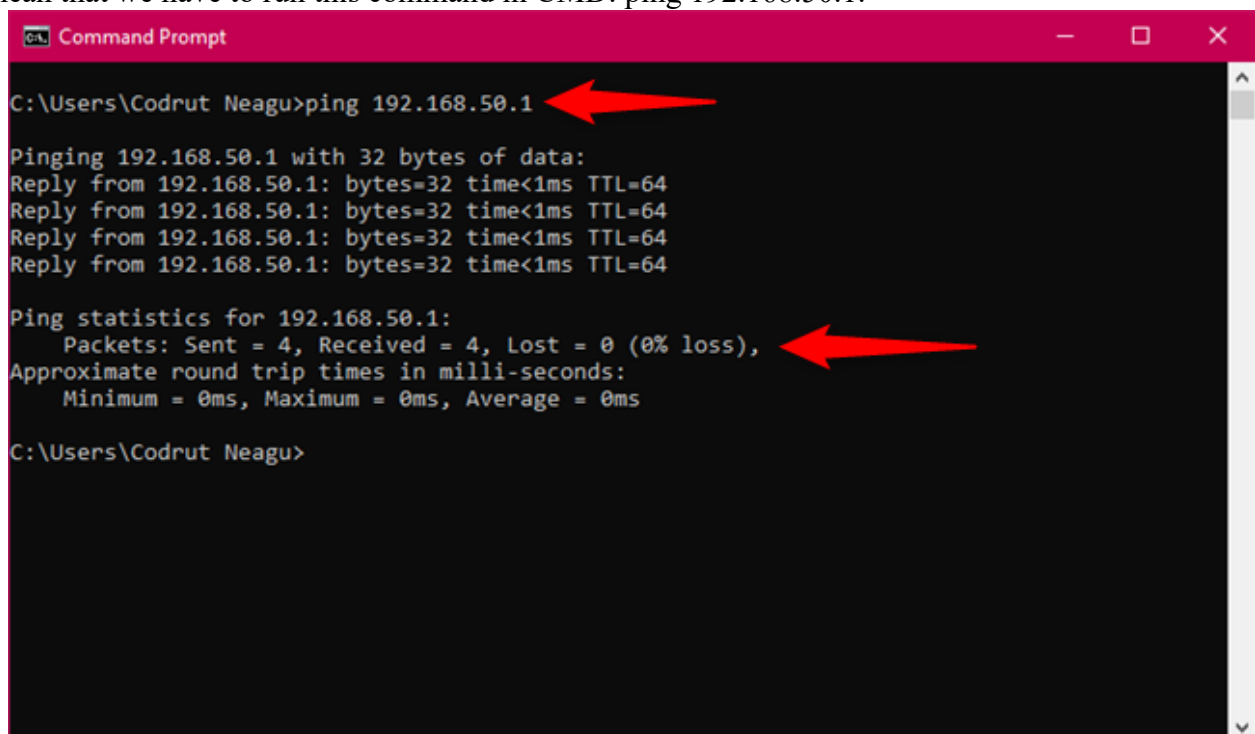
Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected

```

Figure 3: Running ipconfig to identify the IP address of the router

The next step is to check that the network connection between the router and the computer is OK. To do that, it's enough to run the ping command on the router's IP address. In our example, that would mean that we have to run this command in CMD: ping 192.168.50.1.



```

C:\Users\Codrut Neagu>ping 192.168.50.1

Pinging 192.168.50.1 with 32 bytes of data:
Reply from 192.168.50.1: bytes=32 time<1ms TTL=64
Reply from 192.168.50.1: bytes=32 time<1ms TTL=64
Reply from 192.168.50.1: bytes=32 time<1ms TTL=64
Reply from 192.168.50.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Codrut Neagu>

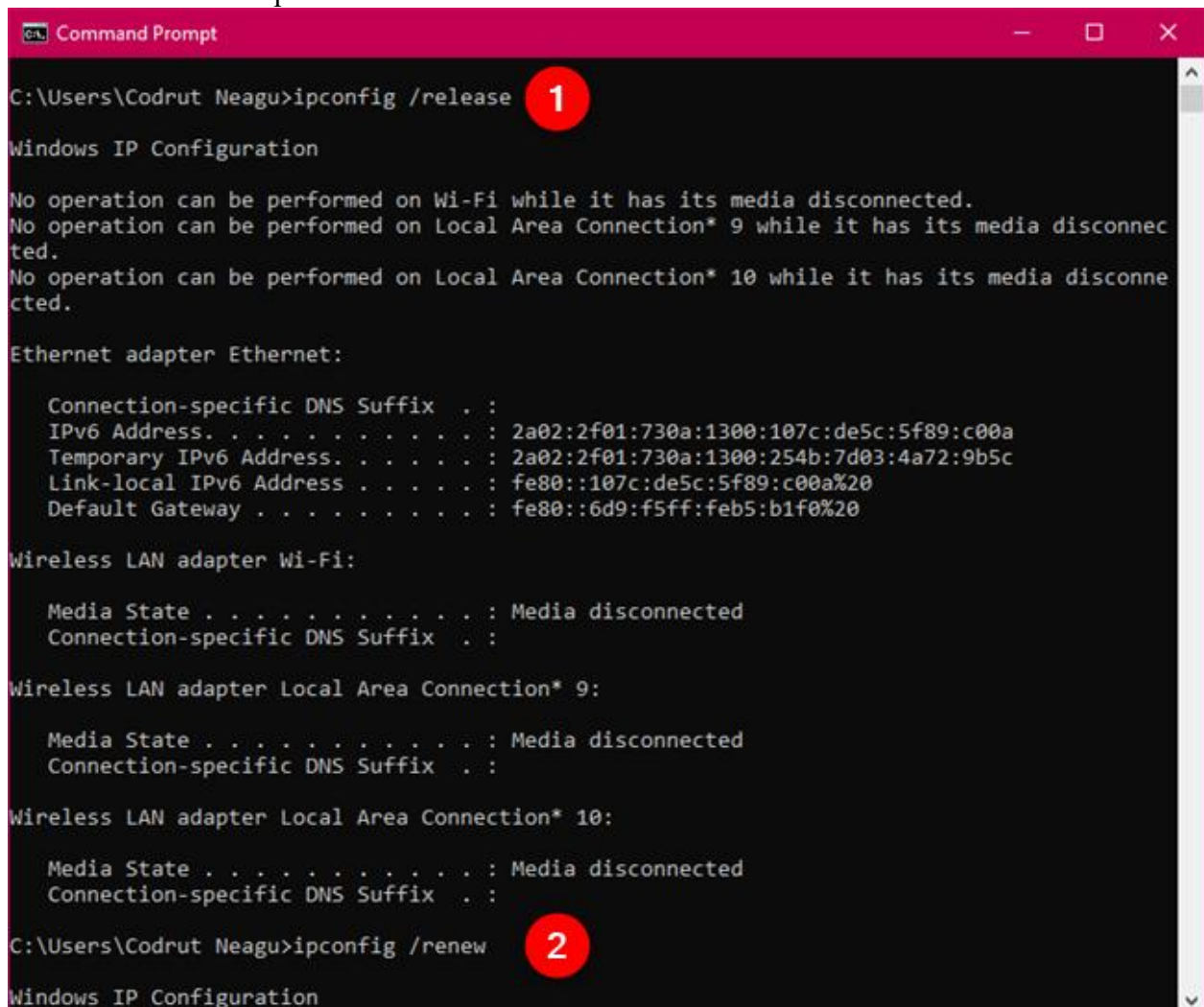
```

Figure4: Pinging the router to check the network connection

If there are no packets lost, then the network connection tested is running well. Otherwise, there's a problem somewhere between your computer and the router, in which case you should check that your PC's network adapter is configured correctly, that the Ethernet cable is OK (if you're using a wired connection), and that the router is configured properly.

How to renew the IP address of your network adapter

When your network connection doesn't work as it should, your network adapter might not have the right IP address assigned. A quick way of trying to solve this issue is to renew its IP address and, fortunately, you can do that quickly, straight from the Command Prompt. Open CMD and run the following commands: `ipconfig /release` and `ipconfig /renew`. The first one (`ipconfig /release`) forces your network adapter to drop its assigned IP address, and the second command (`ipconfig /renew`) renews the network adapter's IP address.



```
C:\Users\Codrut Neagu>ipconfig /release

Windows IP Configuration

No operation can be performed on Wi-Fi while it has its media disconnected.
No operation can be performed on Local Area Connection* 9 while it has its media disconnected.
No operation can be performed on Local Area Connection* 10 while it has its media disconnected.

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2a02:2f01:730a:1300:107c:de5c:5f89:c00a
    Temporary IPv6 Address. . . . . : 2a02:2f01:730a:1300:254b:7d03:4a72:9b5c
    Link-local IPv6 Address . . . . . : fe80::107c:de5c:5f89:c00a%20
    Default Gateway . . . . . : fe80::6d9:f5ff:feb5:b1f0%20

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\Codrut Neagu>ipconfig /renew

Windows IP Configuration
```

Figure5: Running `ipconfig /release` and `ipconfig /renew` to reset the IP address

3 TRACERT Command

Tracert command tells you the path a packet takes from your computer to the destination. It will list all the routers from which a packet passes until it reaches its destination.

C:\tracert google.com

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\Administrator>tracert google.com
Tracing route to google.com [72.14.207.99]
over a maximum of 30 hops:
  0  <1 ms    <1 ms    <1 ms    150.0.0.1
  1  <1 ms    <1 ms    <1 ms    ntc.net.pk [202.83.163.105]
  2  29 ms    29 ms    29 ms    ntc.net.pk [202.83.160.129]
  3  24 ms    26 ms    26 ms    gwish.ntc.net.pk [202.83.160.61]
  4  72 ms    231 ms   268 ms    s8-1-0.rwp44d1.pie.net.pk [202.125.155.65]
  5  49 ms    52 ms    49 ms    rwp44.pie.net.pk [202.125.148.133]
  6  80 ms    52 ms    52 ms    pgs2-2.khi77gsrc1.pie.net.pk [202.125.159.45]
  7  51 ms    46 ms    49 ms    g3-0.khi77gsrc1.pie.net.pk [202.125.128.162]
  8  181 ms   181 ms   182 ms    t2a4-p2-3.uk-lon2.eu.bt.net [166.49.209.51]
  9  180 ms   181 ms   185 ms    t2c2-ge7-0.uk-lon2.eu.bt.net [166.49.176.44]
 10  181 ms   181 ms   182 ms    t2c1-ge4-2.uk-lon1.eu.bt.net [166.49.208.61]
 11  185 ms   185 ms   185 ms    t2a1-pe1.uk-lon1.eu.bt.net [166.49.135.98]
 12  183 ms   185 ms   221 ms    195.66.226.125
 13  179 ms   182 ms   185 ms    72.14.238.242
 14  271 ms   267 ms   268 ms    72.14.236.216
 15  281 ms   268 ms   268 ms    72.14.236.213
 16  326 ms   274 ms   271 ms    72.14.233.115
 17  272 ms   281 ms   304 ms    66.249.94.96
 18  272 ms   277 ms   274 ms    66.249.94.118
 19  275 ms   274 ms   274 ms    72.14.207.99
 20  275 ms   274 ms   274 ms    72.14.207.99

Trace complete.
C:\Documents and Settings\Administrator>
```

4. NSLOOKUP Command

Displays the default DNS server information.

Type the following command

```
C:\>nslookup
```

What is your default DNS server's IP address?

5. NETSTAT Command

You can get other useful cmd nic info from the netstat command, which lets you see the network connections that are active between your system and any other systems on your network or the internet.

Displays active TCP and UDP connections.

Practice the following commands

```
C:\>netstat
```

```
C:\>netstat -a
```

```
C:\>netstat -an
```

```
Command Prompt - netstat
Microsoft Windows [Version 10.0.18363.592]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\Codrut Neagu>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:9012           Codrut-PC:49999        ESTABLISHED
TCP    127.0.0.1:9013           Codrut-PC:50162        ESTABLISHED
TCP    127.0.0.1:9487           Codrut-PC:49815        ESTABLISHED
TCP    127.0.0.1:49815         Codrut-PC:9487         ESTABLISHED
TCP    127.0.0.1:49856         Codrut-PC:49857        ESTABLISHED
TCP    127.0.0.1:49857         Codrut-PC:49856        ESTABLISHED
TCP    127.0.0.1:49860         Codrut-PC:49861        ESTABLISHED
TCP    127.0.0.1:49861         Codrut-PC:49860        ESTABLISHED
TCP    127.0.0.1:49870         Codrut-PC:49871        ESTABLISHED
TCP    127.0.0.1:49871         Codrut-PC:49870        ESTABLISHED
TCP    127.0.0.1:49872         Codrut-PC:49873        ESTABLISHED
TCP    127.0.0.1:49873         Codrut-PC:49872        ESTABLISHED
TCP    127.0.0.1:49876         Codrut-PC:49877        ESTABLISHED
TCP    127.0.0.1:49877         Codrut-PC:49876        ESTABLISHED
TCP    127.0.0.1:49999         Codrut-PC:9012         ESTABLISHED
TCP    127.0.0.1:50014         Codrut-PC:65001        ESTABLISHED
TCP    127.0.0.1:50030         Codrut-PC:50101        ESTABLISHED
TCP    127.0.0.1:50101         Codrut-PC:50030        ESTABLISHED
TCP    127.0.0.1:50162         Codrut-PC:9013         ESTABLISHED
TCP    127.0.0.1:56854         Codrut-PC:56855        ESTABLISHED
TCP    127.0.0.1:56855         Codrut-PC:56854        ESTABLISHED
TCP    127.0.0.1:56859         Codrut-PC:56860        ESTABLISHED
TCP    127.0.0.1:56860         Codrut-PC:56859        ESTABLISHED
TCP    127.0.0.1:57015         Codrut-PC:57016        ESTABLISHED
TCP    127.0.0.1:57016         Codrut-PC:57015        ESTABLISHED
TCP    127.0.0.1:57607         Codrut-PC:57608        ESTABLISHED
TCP    127.0.0.1:57608         Codrut-PC:57607        ESTABLISHED
TCP    127.0.0.1:57692         Codrut-PC:57693        ESTABLISHED
TCP    127.0.0.1:57693         Codrut-PC:57692        ESTABLISHED
TCP    127.0.0.1:65001         Codrut-PC:50014        ESTABLISHED
TCP    192.168.50.239:58685     51.105.249.228:https   ESTABLISHED
TCP    192.168.50.239:58692     ec2-54-190-34-249:https ESTABLISHED
TCP    192.168.50.239:58696     136:http               ESTABLISHED
TCP    192.168.50.239:58706     51.105.249.228:https   ESTABLISHED
TCP    192.168.50.239:58750     ec2-3-120-198-117:https ESTABLISHED
TCP    192.168.50.239:59957     53:https               ESTABLISHED
TCP    192.168.50.239:60094     do-1:https             ESTABLISHED
```

Netstat shows the active network connections and open ports

If you add the -a parameter to the netstat command, you can get a list with all the connections and listening ports, as seen in the image below.

```
Command Prompt - netstat -a

C:\Users\Codrut Neagu>netstat -a

Active Connections

Proto Local Address Foreign Address State
TCP 0.0.0.0:135 Codrut-PC:0 LISTENING
TCP 0.0.0.0:445 Codrut-PC:0 LISTENING
TCP 0.0.0.0:902 Codrut-PC:0 LISTENING
TCP 0.0.0.0:912 Codrut-PC:0 LISTENING
TCP 0.0.0.0:5040 Codrut-PC:0 LISTENING
TCP 0.0.0.0:5357 Codrut-PC:0 LISTENING
TCP 0.0.0.0:7680 Codrut-PC:0 LISTENING
TCP 0.0.0.0:9012 Codrut-PC:0 LISTENING
TCP 0.0.0.0:9013 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49664 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49665 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49666 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49667 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49670 Codrut-PC:0 LISTENING
TCP 0.0.0.0:49844 Codrut-PC:0 LISTENING
TCP 0.0.0.0:57621 Codrut-PC:0 LISTENING
TCP 0.0.0.0:61688 Codrut-PC:0 LISTENING
TCP 127.0.0.1:1042 Codrut-PC:0 LISTENING
TCP 127.0.0.1:1043 Codrut-PC:0 LISTENING
TCP 127.0.0.1:3213 Codrut-PC:0 LISTENING
TCP 127.0.0.1:9012 Codrut-PC:49999 ESTABLISHED
TCP 127.0.0.1:9013 Codrut-PC:50162 ESTABLISHED
TCP 127.0.0.1:9487 Codrut-PC:0 LISTENING
TCP 127.0.0.1:9487 Codrut-PC:49815 ESTABLISHED
TCP 127.0.0.1:13010 Codrut-PC:0 LISTENING
TCP 127.0.0.1:13030 Codrut-PC:0 LISTENING
TCP 127.0.0.1:17945 Codrut-PC:0 LISTENING
TCP 127.0.0.1:49815 Codrut-PC:9487 ESTABLISHED
TCP 127.0.0.1:49856 Codrut-PC:49857 ESTABLISHED
TCP 127.0.0.1:49857 Codrut-PC:49856 ESTABLISHED
TCP 127.0.0.1:49860 Codrut-PC:49861 ESTABLISHED
TCP 127.0.0.1:49861 Codrut-PC:49860 ESTABLISHED
TCP 127.0.0.1:49870 Codrut-PC:49871 ESTABLISHED
TCP 127.0.0.1:49871 Codrut-PC:49870 ESTABLISHED
TCP 127.0.0.1:49872 Codrut-PC:49873 ESTABLISHED
TCP 127.0.0.1:49873 Codrut-PC:49872 ESTABLISHED
TCP 127.0.0.1:49876 Codrut-PC:49877 ESTABLISHED
TCP 127.0.0.1:49877 Codrut-PC:49876 ESTABLISHED
TCP 127.0.0.1:49999 Codrut-PC:9012 ESTABLISHED
TCP 127.0.0.1:50014 Codrut-PC:65001 ESTABLISHED
TCP 127.0.0.1:50030 Codrut-PC:0 LISTENING
TCP 127.0.0.1:50030 Codrut-PC:50101 ESTABLISHED
TCP 127.0.0.1:50101 Codrut-PC:50030 ESTABLISHED
TCP 127.0.0.1:50162 Codrut-PC:9013 ESTABLISHED
TCP 127.0.0.1:56854 Codrut-PC:56855 ESTABLISHED
TCP 127.0.0.1:56855 Codrut-PC:56854 ESTABLISHED
TCP 127.0.0.1:56859 Codrut-PC:56860 ESTABLISHED
TCP 127.0.0.1:56860 Codrut-PC:56859 ESTABLISHED
TCP 127.0.0.1:57015 Codrut-PC:57016 ESTABLISHED
TCP 127.0.0.1:57016 Codrut-PC:57015 ESTABLISHED
TCP 127.0.0.1:57607 Codrut-PC:57608 ESTABLISHED
TCP 127.0.0.1:57608 Codrut-PC:57607 ESTABLISHED
TCP 127.0.0.1:57692 Codrut-PC:57693 ESTABLISHED
TCP 127.0.0.1:57693 Codrut-PC:57692 ESTABLISHED
TCP 127.0.0.1:65001 Codrut-PC:0 LISTENING
TCP 127.0.0.1:65001 Codrut-PC:50014 ESTABLISHED
TCP 192.168.50.239:139 Codrut-PC:0 LISTENING
TCP 192.168.50.239:58685 51.105.249.228:https ESTABLISHED
```

Netstat -a displays the active network connections, open ports and listening ports

6. ARP Command

ARP command corresponds to the Address Resolution Protocol, it is easy to understand of network communications in terms of IP addressing, packet delivery is ultimately dependent on the Media Access Control (MAC) address of the device's network adapter. This is where the Address Resolution Protocol comes into play. Its job is to map IP addresses to MAC addresses.

Windows devices maintain an ARP cache, which contains the results of recent ARP queries. It shows the contents of this cache by using the ARP -A command. If any problems in communicating with one specific host, you can append the remote host's IP address to the ARP -A command.

```
Command Prompt
C:\Users\Codrut Neagu>arp -a

Interface: 192.168.50.239 --- 0x14
Internet Address      Physical Address      Type
192.168.50.1          04-d9-f5-b5-b1-f0    dynamic
192.168.50.4          90-94-97-c6-42-62    dynamic
192.168.50.60         ac-e4-b5-e0-5c-57    dynamic
192.168.50.96         00-28-f8-3c-eb-5a    dynamic
192.168.50.169        54-25-ea-a0-0a-51    dynamic
192.168.50.205        04-d9-f5-b5-dd-10    dynamic
192.168.50.247        b0-6e-bf-10-d1-62    dynamic
192.168.50.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\Codrut Neagu>
```

```
C:\WINDOWS\system32\cmd.exe
C:\Users\Brien>arp -a 147.100.100.151

Interface: 147.100.100.224 --- 0x2
Internet Address      Physical Address      Type
147.100.100.151      68-05-ca-19-1c-d2    dynamic

C:\Users\Brien>
```

7.NbtStat-n Command

The NbtStat -n command for example, shows the NetBIOS names that are in use by a device. The NbtStat -r command shows how many NetBIOS names the device has been able to resolve recently.

8.Route Command

IP networks use routing tables to direct packets from one subnet to another. The Windows Route utility allows you to view the device's routing tables. The Route command is that it not only shows you the routing table, it lets you make changes. Commands such as Route Add, Route Delete, and Route Change allow you to make routing table modifications on an as needed basis.

```
Command Prompt
C:\Users\Admin>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric] [IF interface]

-f          Clears the routing tables of all gateway entries. If this is
            used in conjunction with one of the commands, the tables are
            cleared prior to running the command.

-p          When used with the ADD command, makes a route persistent across
            boots of the system. By default, routes are not preserved
            when the system is restarted. Ignored for all other commands,
            which always affect the appropriate persistent routes.

-4          Force using IPv4.

-6          Force using IPv6.

command    One of these:
```

9. GETMAC Command

Getmac is a Windows command used to display the Media Access Control (MAC) addresses for each network adapter in the computer. One of the fastest and easiest ways to obtain the MAC addresses of your network adapters is to use the getmac command. In Command Prompt, type getmac and press Enter, as seen in the image below.

```
Command Prompt
C:\Users\Codrut Neagu>getmac

Physical Address      Transport Name
-----
04-D9-F5-34-B1-A3     \Device\Tcpip_{94EBAE5D-9C8B-40A3-90CF-7D806D1D0F4D}
04-D9-F5-34-B1-A2     Media disconnected
38-00-25-41-C3-F5      Media disconnected
00-50-56-C0-00-01     \Device\Tcpip_{1B88E222-B53F-42AF-9029-79400A0F750F}
00-50-56-C0-00-08     \Device\Tcpip_{7F80AD46-E70A-4A38-877E-02763783EB92}

C:\Users\Codrut Neagu>
```

10.SYSTEMINFO Command: System Information

If you need to know what brand of network card you have, processor details, or the exact version of your Windows OS, the SYSTEMINFO command can help. This command polls your system and pulls the most important information about your system. It lists the information in a clean format that's easy to read.

```
Command Prompt
Microsoft Windows [Version 10.0.18362.959]
(c) 2019 Microsoft Corporation. All rights reserved.
C:\Users\Admin>systeminfo

Host Name:                  DESKTOP-ETD1J12
OS Name:                    Microsoft Windows 10 Pro
OS Version:                 10.0.18362.959 Build 18362
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:               Multiprocessor Free
Registered Owner:           Windows User
Registered Organization:
Product ID:                  00331-10000-00001-AA797
Original Install Date:       2/2/2020, 1:00:24 PM
System Boot Time:            8/10/2020, 9:06:30 AM
System Manufacturer:        HP
System Model:                HP 240 G6 Notebook PC
System Type:                 x64-based PC
Processor(s):                1 Processor(s) Installed.
                             [01]: Intel(R) Family 6 Model 78 Stepping 3 GenuineIntel ~1992 Mhz
BIOS Version:                Insyde F.32, 2/26/2018
Windows Directory:           C:\Windows
System Directory:            C:\Windows\System32
Boot Device:                 \Device\HarddiskVolume2
System Locale:                en-us;English (United States)
Input Locale:                en-us;English (United States)
Time Zone:                   (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:       3,982 MB
Available Physical Memory:   227 MB
Virtual Memory: Max Size:    9,741 MB
Virtual Memory: Available:   4,825 MB
Virtual Memory: In Use:      3,716 MB
Page File Location(s):       C:\pagefile.sys
Domain:                       \DESKTOP-ETD1J12
Logon Server:
Network(s):                  [01]: KB4565623
                             [02]: KB4502166
                             [03]: KB4498523
                             [04]: KB4503368
                             [05]: KB4537759
                             [06]: KB4538674
                             [07]: KB4541158

Activate Windows
Go to Settings to activate Windows.
```

1(b) Identify and explore Network devices, models and cables

1. Aim

To identify and understand the functions of various network devices, reference models, transmission media, and connectors used in computer networks.

2. Learning Objectives

After completing this experiment, students will be able to:

- Identify common networking devices.
- Explain the functions of network devices.
- Differentiate between the OSI and TCP/IP reference models.
- Identify various transmission media and network cables.
- Select appropriate networking devices and cables for different applications.

3. Introduction

A **computer network** is a collection of interconnected devices that communicate with each other to exchange data and share resources. Networking devices and communication media play an essential role in ensuring efficient, secure, and reliable communication.

4. Network Devices

4.1 Router: A **router** is a Layer 3 networking device that connects multiple networks and forwards packets based on IP addresses.

Functions

- Connects different networks.
- Determines the best path for packet transmission.
- Provides Internet connectivity.
- Supports NAT and DHCP.

Applications

- Home Wi-Fi Router
- Enterprise Networks
- ISP Networks

4.2 Switch: A **switch** is a Layer 2 networking device that connects devices within a Local Area Network (LAN).

Functions

- Learns MAC addresses.
- Sends frames only to the intended destination.
- Reduces network collisions.
- Improves network performance.

Applications

- Offices
- Computer Labs
- Data Centers

4.3 Hub: A **hub** is a basic networking device that broadcasts received data to all connected devices.

Characteristics

- Layer 1 Device
- No MAC address learning
- Less secure
- Causes collisions

4.4 Bridge: A bridge connects two LAN segments and filters traffic using MAC addresses.

Advantages

- Reduces collisions
- Improves network efficiency

4.5 Repeater: A repeater regenerates weak signals to extend communication distance.

Applications

- Long-distance Ethernet
- Fiber optic communication

4.6 Gateway: A gateway connects networks using different communication protocols.

Example

Connecting an IPv4 network with an IPv6 network.

4.7 Modem: A modem converts digital signals into analog signals and vice versa.

Types

- DSL Modem
- Cable Modem
- Fiber Modem

4.8 Wireless Access Point (WAP): Provides wireless connectivity to users in a network.

Features

- Wi-Fi connectivity
- Supports multiple users
- Extends wireless coverage

4.9 Network Interface Card (NIC): A hardware component that enables computers to connect to a network.

Types

- Wired NIC
- Wireless NIC

5. Reference Models

5.1 OSI Reference Model

The Open Systems Interconnection (OSI) model consists of **seven layers**.

Layer	Name	Function
7	Application	User Interface
6	Presentation	Encryption & Compression
5	Session	Session Management
4	Transport	Reliable Communication
3	Network	Routing
2	Data Link	Framing & MAC Address
1	Physical	Transmission of Bits

Advantages

- Standardized communication
- Easy troubleshooting
- Modular architecture

5.2 TCP/IP Model

The TCP/IP model consists of **four layers**.

Layer	Function
Application	HTTP, FTP, SMTP
Transport	TCP, UDP
Internet	IP Routing
Network Access	Ethernet, Wi-Fi

OSI vs TCP/IP

OSI	TCP/IP
7 Layers	4 Layers
Theoretical Model	Practical Model
ISO Standard	Internet Standard

6. Transmission Media

Transmission media are used to carry data from one device to another.

6.1 Guided Media:

Twisted Pair Cable (UTP)

Advantages

- Low cost
- Easy installation

Disadvantages

- Limited distance
- Susceptible to interference

Applications

- LAN

Shielded Twisted Pair (STP)

Advantages

- Better noise protection

Applications

- Industrial Networks

Coaxial Cable

Applications

- Cable TV
- CCTV

Advantages

- Better shielding

Fiber Optic Cable

Advantages

- High bandwidth
- Long-distance communication
- Secure

Disadvantages

- Expensive

Applications

- Internet Backbone
- Data Centers

6.2 Unguided Media

- Radio Waves
- Microwaves
- Infrared
- Satellite Communication

7. Connectors

RJ45 Connector

Used for Ethernet communication.

Characteristics

- 8 Pins
- Used with Cat5e/Cat6/Cat6A cables
- Supports Gigabit Ethernet

8. Comparison of Network Devices

Device	Layer	Address Used	Main Function
Hub	1	None	Broadcasts Data
Switch	2	MAC	Connects LAN Devices
Bridge	2	MAC	Connects LAN Segments
Router	3	IP	Routes Packets
Gateway	Multiple	Multiple	Protocol Conversion
Repeater	1	None	Signal Regeneration
Access Point	2	MAC	Wireless Connectivity
Modem	1	None	Signal Conversion

9. Applications

- Home Networks
- Office Networks
- Cloud Computing
- IoT Networks
- Data Centers
- Campus Networks
- Smart Cities
- Industrial Automation

10. Advantages

- Efficient communication
- Resource sharing
- Secure connectivity
- High-speed data transfer
- Scalable network infrastructure

11. Conclusion

This experiment provides an understanding of the **fundamental networking devices, reference models, transmission media, and connectors** used in modern computer networks. Identifying the appropriate device and communication medium is essential for designing efficient, scalable, and reliable network infrastructures. These concepts form the foundation for advanced topics such as routing, switching, wireless communication, network security, and cloud networking.

Experiment 2

2(a) Introduction to Cisco packet tracer.

2(b) Create various network topologies using a cisco packet tracer.

CISCO PACKET TRACER TOOL

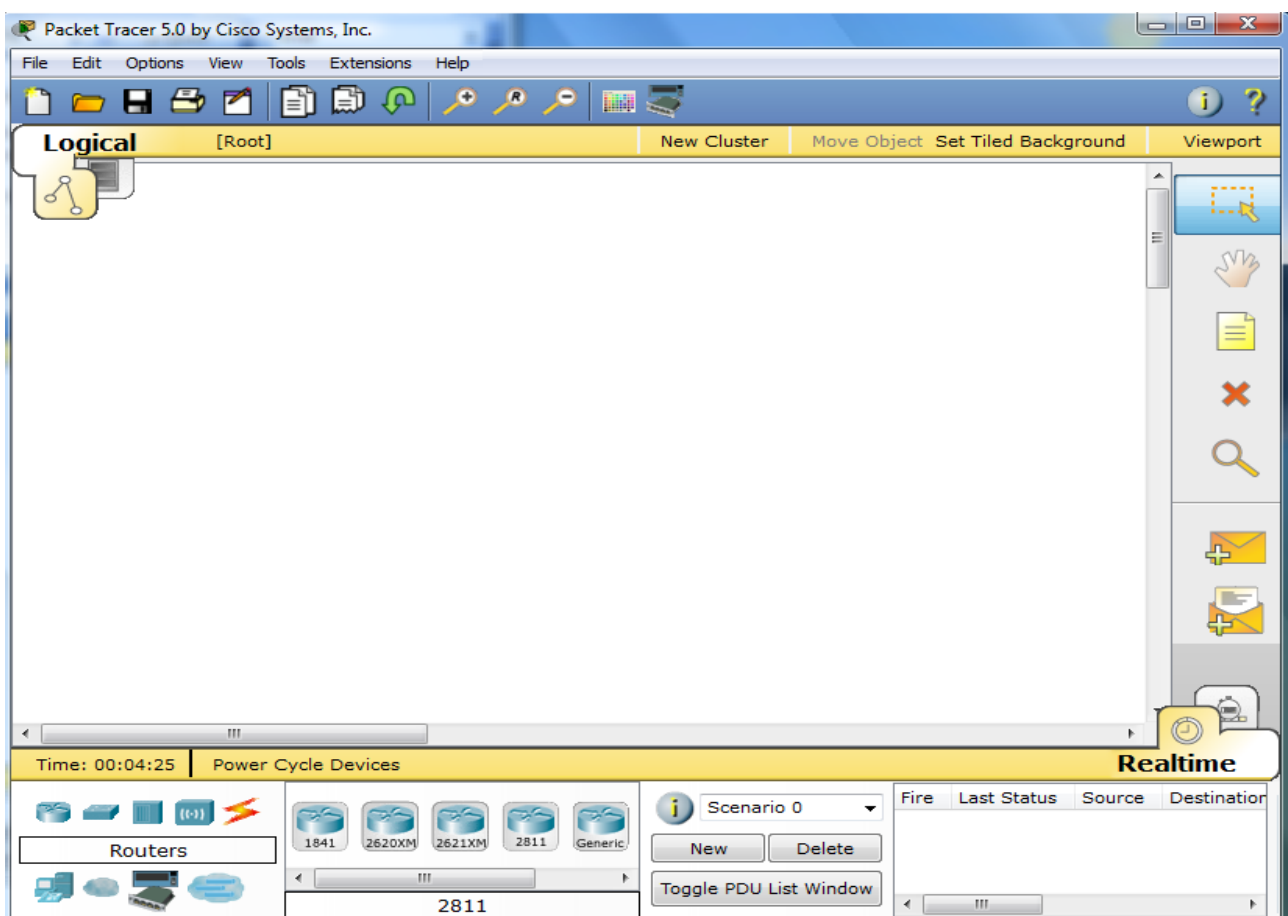
Packet Tracer: Creating a New Topology

What is Packet Tracer? Packet Tracer is a protocol simulator developed by Dennis Frezzo and his team at Cisco Systems. Packet Tracer (PT) is a powerful and dynamic tool that displays the various protocols used in networking, in either Real Time or Simulation mode. This includes layer 2 protocols such as Ethernet and PPP, layer 3 protocols such as IP, ICMP, and ARP, and layer 4 protocols such as TCP and UDP. Routing protocols can also be traced.

Purpose: The purpose of this lab is to become familiar with building topologies in Packet Tracer.

Version: This lab is based on Packet Tracer 5.0, 7.3.0

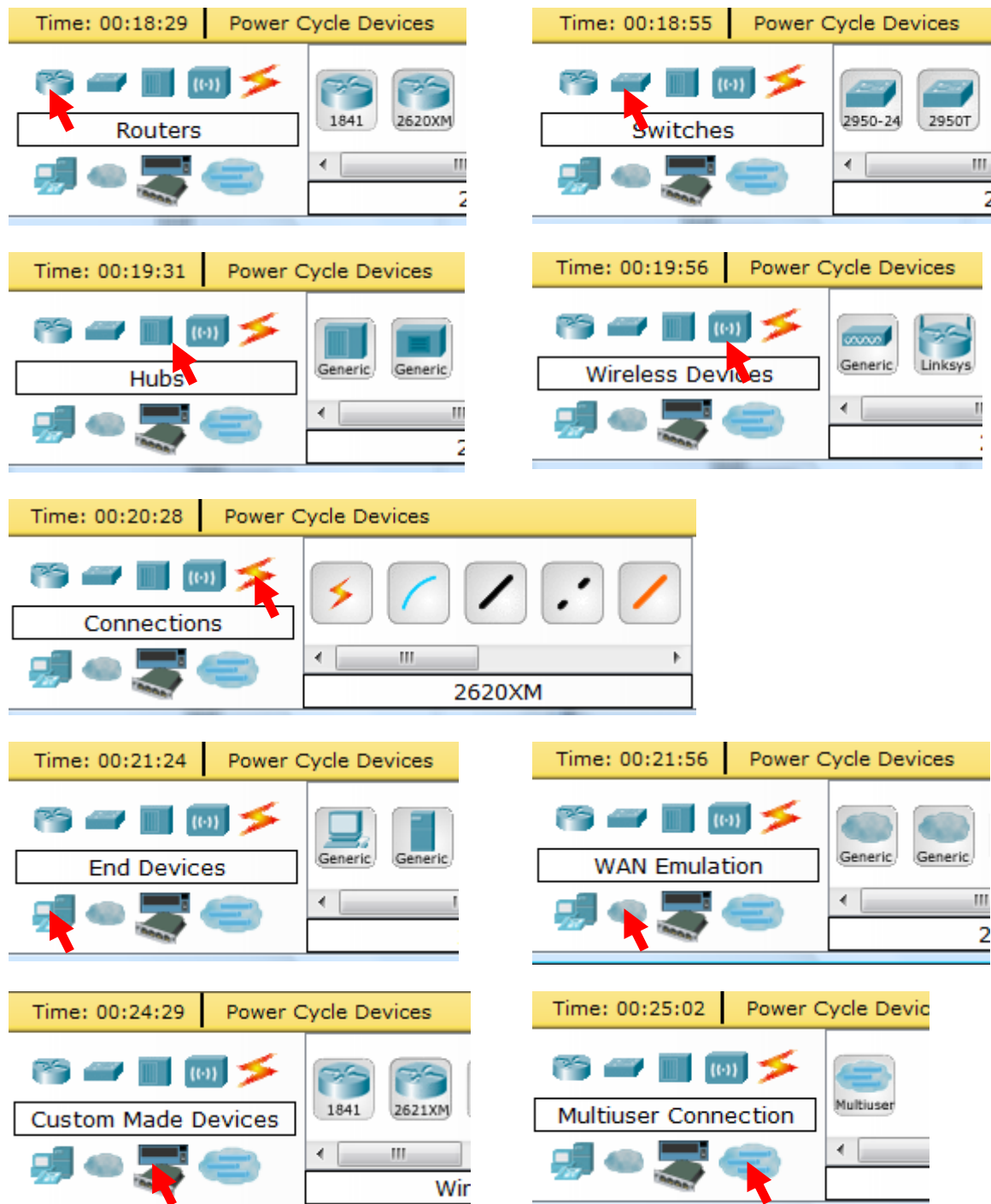
Step 1: Start Packet Tracer



Step 2: Choosing Devices and Connections

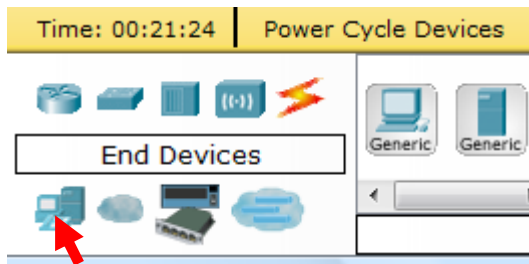
We will begin building our network topology by selecting devices and the media in which to connect them. Several types of devices and network connections can be used. For this lab we will keep it simple by using **End Devices**, **Switches**, **Hubs**, and **Connections**.

Single click on each group of devices and connections to display the various choices. The devices you see may differ slightly.

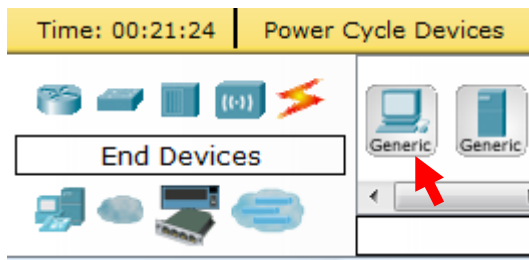


Step 3: Building the Topology -Adding Hosts

Single click on the **End Devices**.



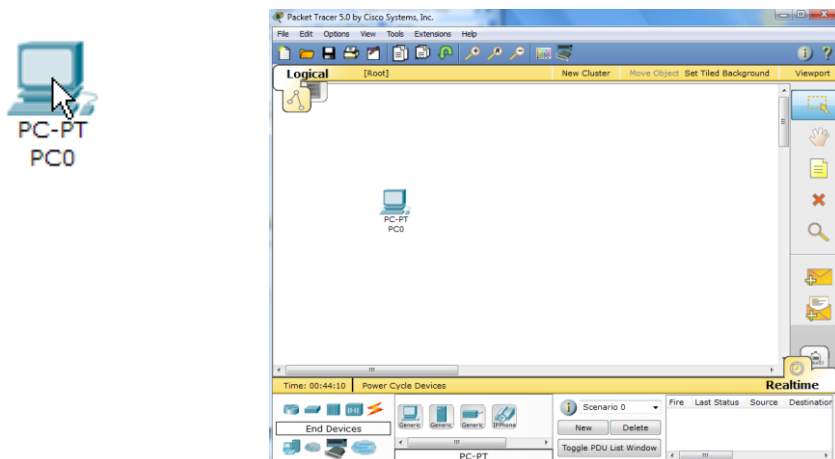
Single click on the **Generic** host.



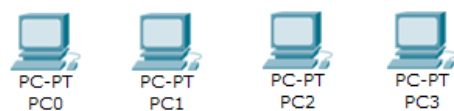
Move the cursor into topology area. You will notice it turns into a plus “+” sign.

+

Single click in the topology area and it copies the device.



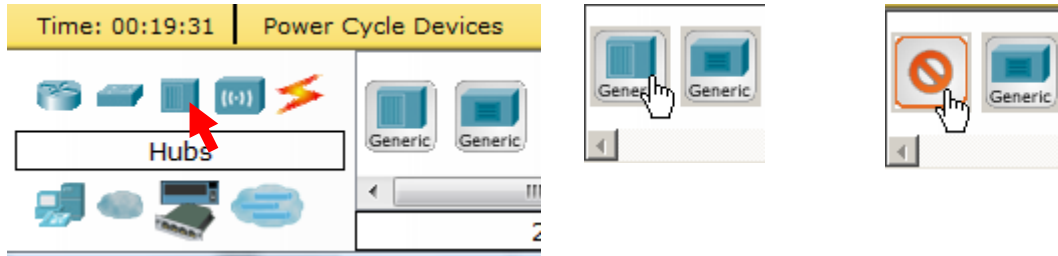
Add three more hosts.



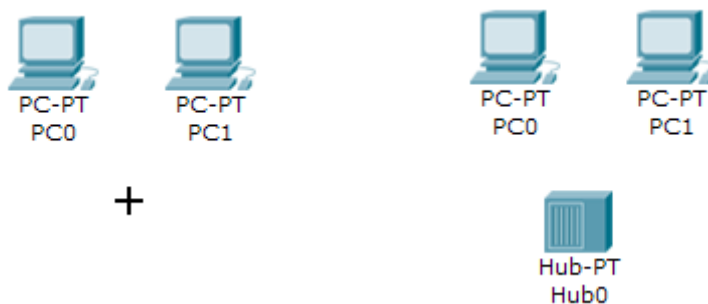
Step 4: Building the Topology - Connecting the Hosts to Hubs and Switches

Adding a Hub

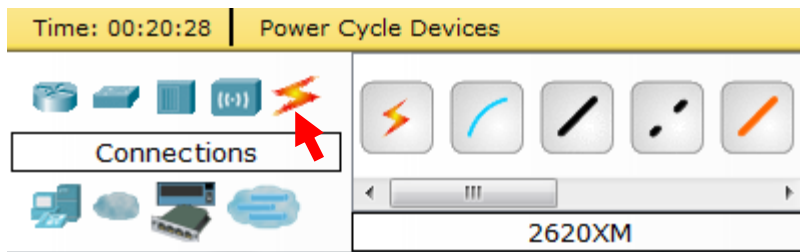
Select a hub, by clicking once on **Hubs** and once on a **Generic** hub.



Add the hub by moving the plus sign “+” below PC0 and PC1 and click once.



Connect PC0 to Hub0 by first choosing **Connections**.



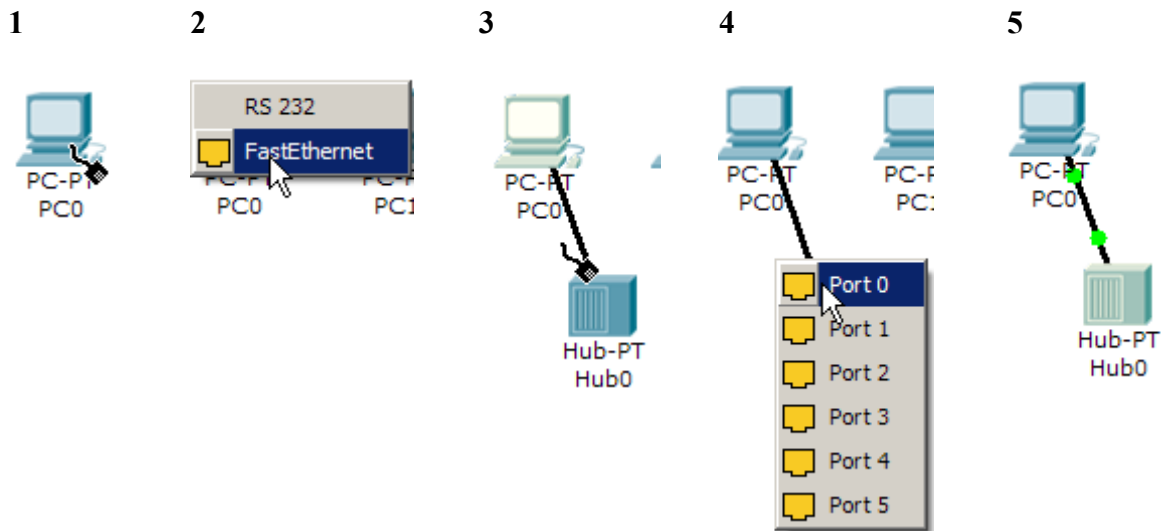
Click once on the **Copper Straight-through** cable.



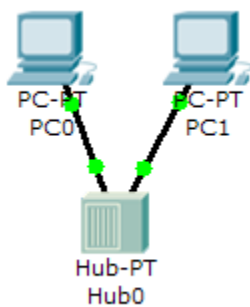
Perform the following steps to connect **PC0** to **Hub0**:

1. Click once on **PC0**
2. Choose **FastEthernet**
3. Drag the cursor to **Hub0**
4. Click once on **Hub0** and choose **Port 0**

5. Notice the green link lights on both the **PC0** Ethernet NIC and the **Hub0** Port 0 showing that the link is active.

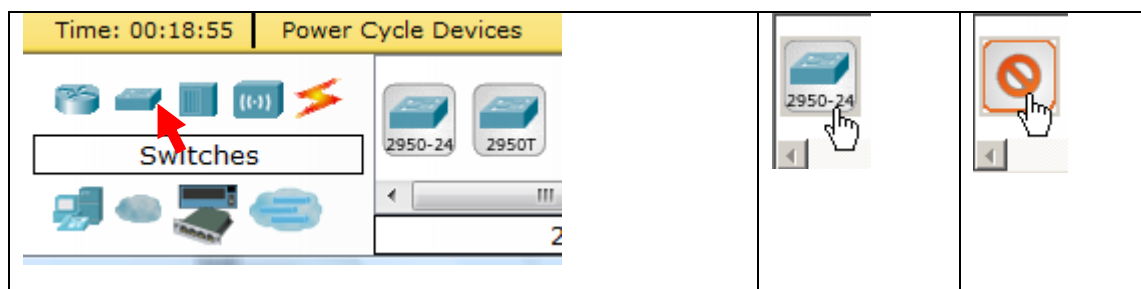


Repeat the steps above for **PC1** connecting it to **Port 1** on **Hub0**. (The actual hub port you choose does not matter.)

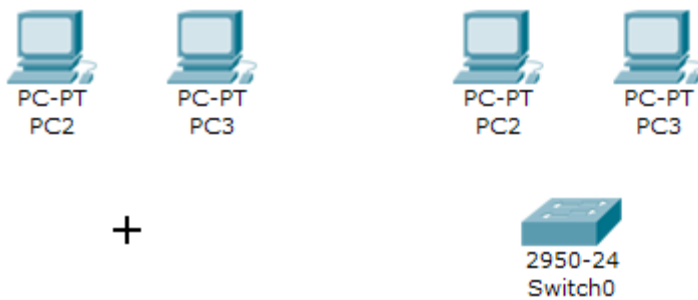


Adding a Switch

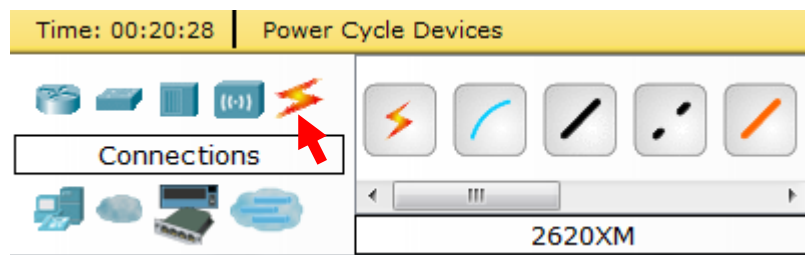
Select a switch, by clicking once on **Switches** and once on a **2950-24** switch.



Add the switch by moving the plus sign “+” below PC2 and PC3 and click once.



Connect PC2 to Hub0 by first choosing **Connections**.



Click once on the **Copper Straight-through** cable.

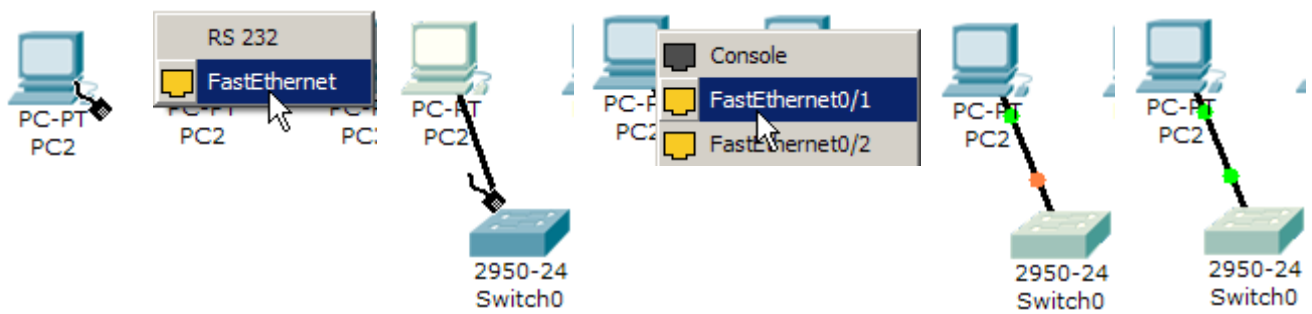


Perform the following steps to connect **PC2** to **Switch0**:

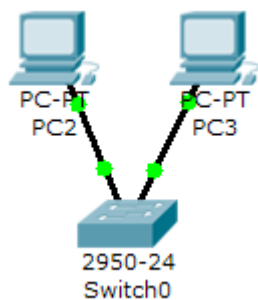
1. Click once on **PC2**
2. Choose **FastEthernet**
3. Drag the cursor to **Switch0**
4. Click once on **Switch0** and choose **FastEthernet0/1**
5. Notice the green link lights on **PC2** Ethernet NIC and amber light **Switch0 FastEthernet0/1 port**. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.
6. After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now forwarded out the switch port.

Note: Spanning Tree Protocol (STP) is discussed later.

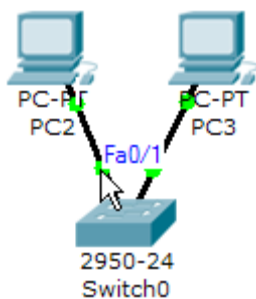
1 2 3 4 5 6



Repeat the steps above for **PC3** connecting it to **Port 3** on **Switch0** on port **FastEtherent0/2**. (The actual switch port you choose does not matter.)



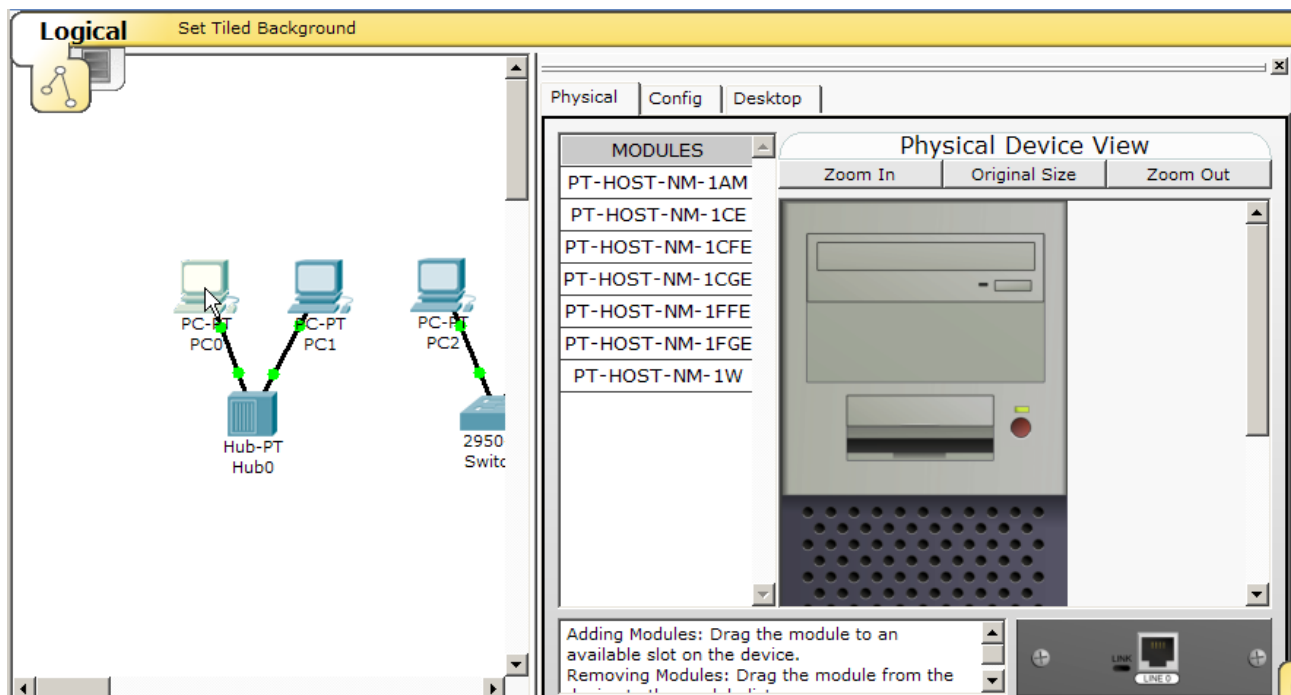
Move the cursor over the link light to view the port number. **Fa** means FastEthernet, 100 Mbps Ethernet.



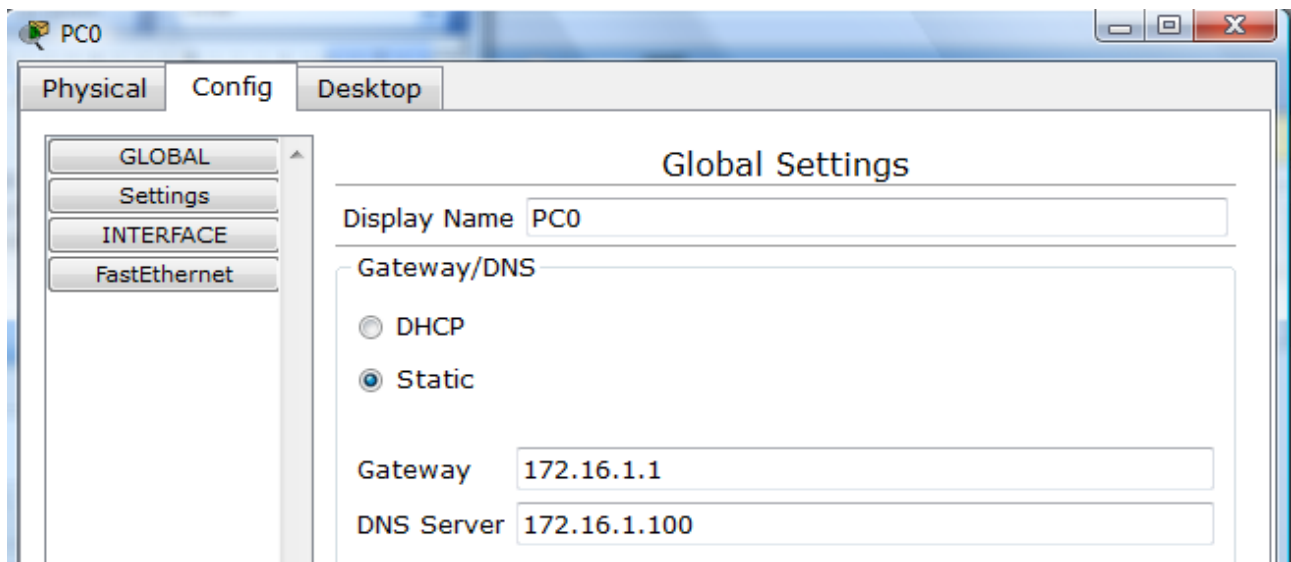
Step 5: Configuring IP Addresses and Subnet Masks on the Hosts

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

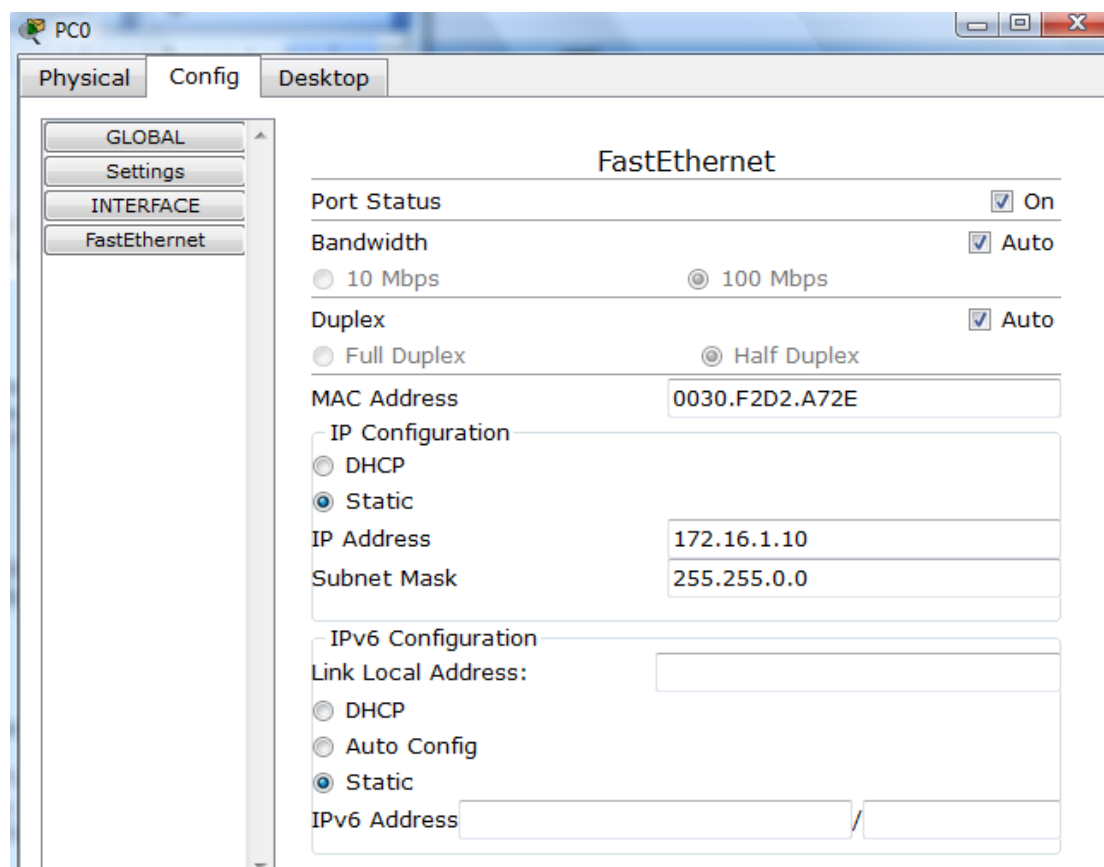
Click once on PC0.



Choose the **Config** tab and click on **Settings**. It is here that you can change the name of PC0. It is also here where you would enter a **Gateway IP Address**, also known as the default gateway and the **DNS Server IP Address**. We will discuss this later, but this would be the IP address of the local router. If you want, you can enter the Gateway IP Address 172.16.1.1 and DNS Server IP Address 172.16.1.100, although it will not be used in this lab.



Click on **Interface** and then **FastEthernet**. Although we have not yet discussed IP Addresses, add the IP Address to 172.16.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.0.0. We will discuss this later.



The screenshot shows a configuration window for PC0 with tabs for Physical, Config, and Desktop. The Config tab is active, and the left sidebar shows a tree view with GLOBAL, Settings, INTERFACE, and FastEthernet. The FastEthernet interface is selected, and its configuration is displayed on the right. The Port Status is set to On. The Bandwidth is set to 100 Mbps (selected) and 10 Mbps (unselected). The Duplex is set to Auto (selected) and Half Duplex (unselected). The MAC Address is 0030.F2D2.A72E. The IP Configuration is set to Static (selected) and DHCP (unselected). The IP Address is 172.16.1.10 and the Subnet Mask is 255.255.0.0. The IPv6 Configuration section is also visible, with Link Local Address, DHCP, Auto Config, and Static options, and an IPv6 Address field.

Also, notice this is where you can change the Bandwidth (speed) and Duplex of the Ethernet NIC (Network Interface Card). The default is Auto (autonegotiation), which means the NIC will negotiate with the hub or switch. The bandwidth and/or duplex can be manually set by removing the check from the **Auto** box and choosing the specific option.

Bandwidth - Auto

If the host is connected to a hub or switch port which can do 100 Mbps, then the Ethernet NIC on the host will choose 100 Mbps (Fast Ethernet). Otherwise, if the hub or switch port can only do 10 Mbps, then the Ethernet NIC on the host will choose 10 Mbps (Ethernet).

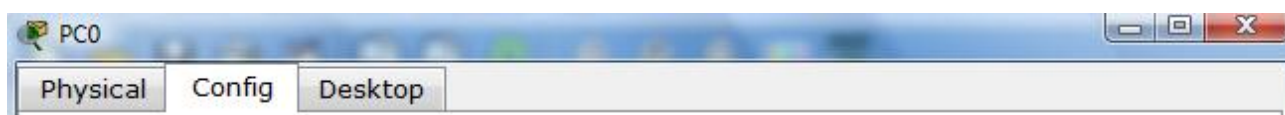
Duplex - Auto

Hub: If the host is connected to a hub, then the Ethernet NIC on the host will choose Half Duplex.

Switch: If the host is connected to a switch, and the switch port is configured as Full Duplex (or Autonegotiation), then the Ethernet NIC on the host will choose Full Duplex. If the switch port is configured as Half Duplex, then the Ethernet NIC on the host will choose Half Duplex. (Full Duplex is a much more efficient option.)

The information is automatically saved when entered.

To close this dialog box, click the “X” in the upper right.

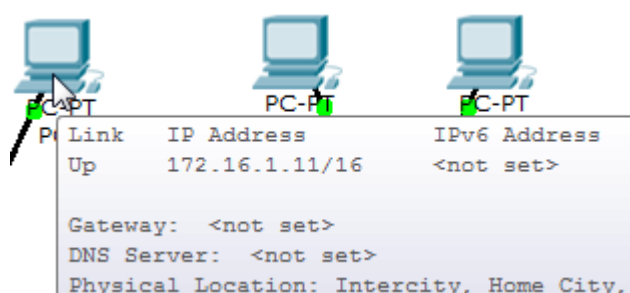


Repeat these steps for the other hosts. Use the information below for IP Addresses and Subnet Masks.

<u>Host</u>	<u>IP Address</u>	<u>Subnet Mask</u>
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

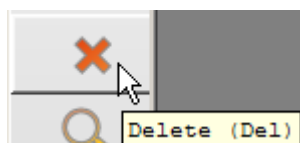
Verify the information

To verify the information that you entered, move the Select tool (arrow) over each host.



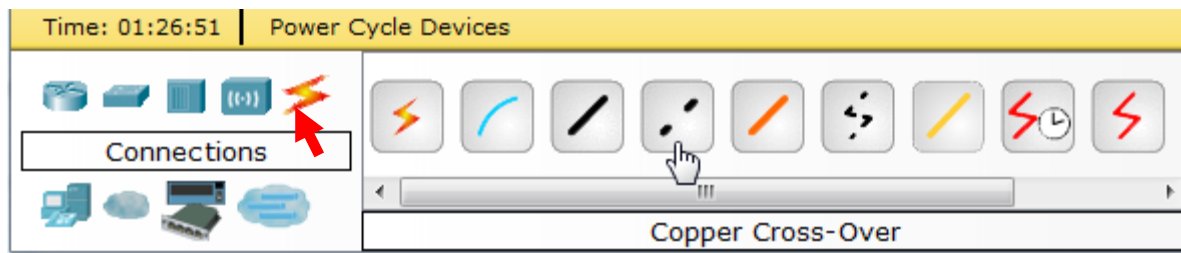
Deleting a Device or Link

To delete a device or link, choose the **Delete** tool and click on the item you wish to delete.

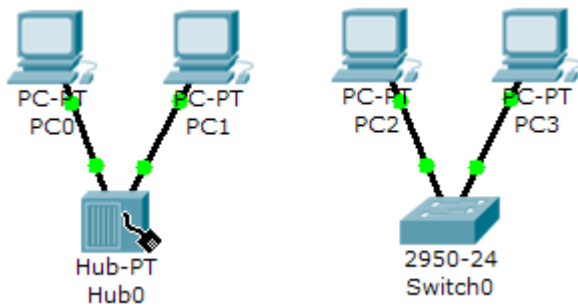


Step 6: Connecting Hub0 to Switch0

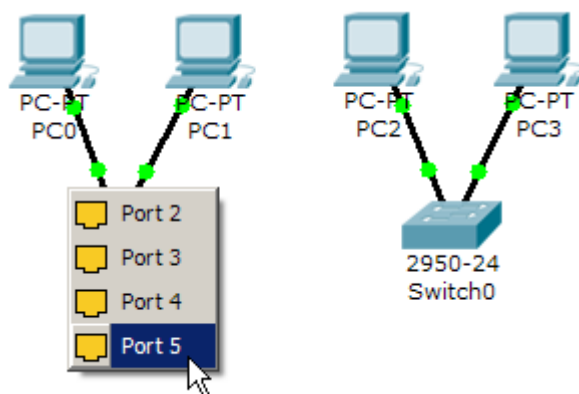
To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable. Click once the **Cross-over** Cable from the **Connections** options.



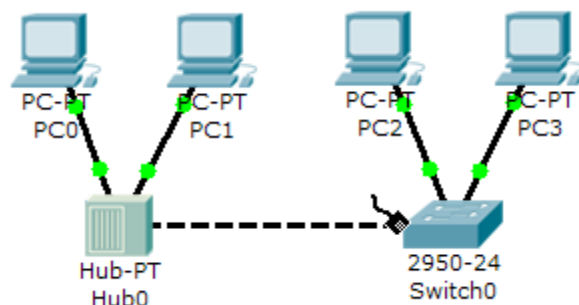
Move the Connections cursor over **Hub0** and click once.



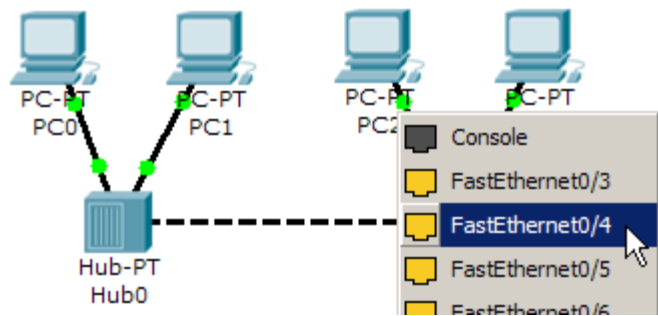
Select **Port 5** (actual port does not matter).



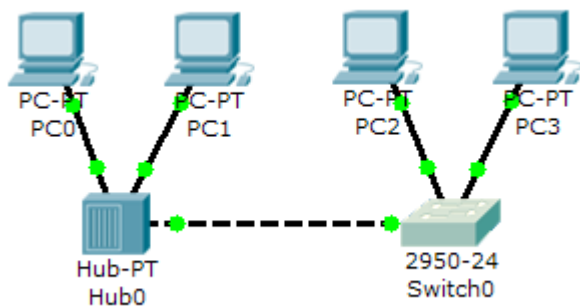
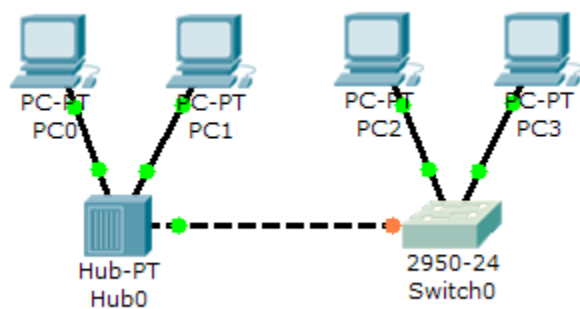
Move the Connections cursor to **Switch0**.



Click once on **Switch0** and choose **FastEthernet0/4** (actual port does not matter).



The link light for switch port **FastEthernet0/4** will begin as amber and eventually change to green as the Spanning Tree Protocol transitions the port to forwarding.



Step 7: Verifying Connectivity in Realtime Mode

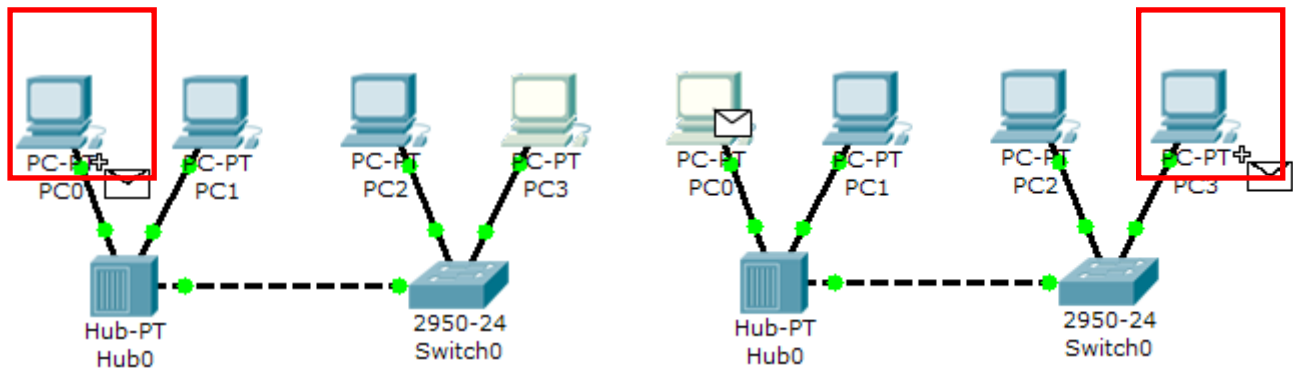
Be sure you are in **Realtime** mode.



Select the **Add Simple PDU** tool used to ping devices.



Click once on PC0, then once on PC3.



The PDU **Last Status** should show as **Successful**.

Realtime				
Fire	Last Status	Source	Destination	Type
	Successful	PC0	PC3	ICMP

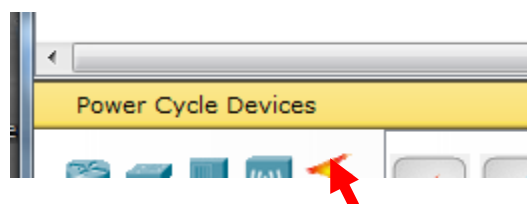
Resetting the Network

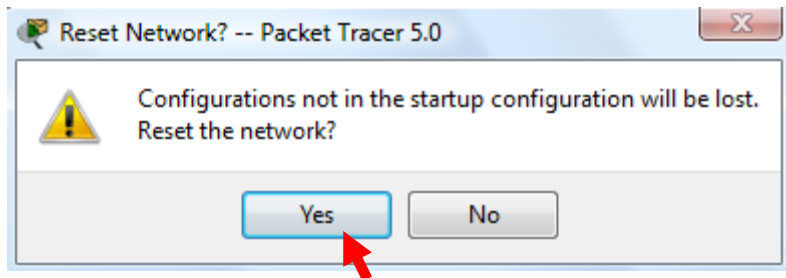
At this point we will want to reset the network, Whenever you want to reset the network and begin the simulation again, perform the following tasks:

Click **Delete** in the PDU area.

<

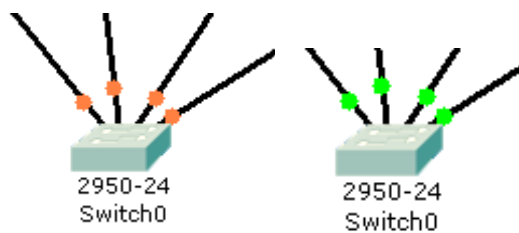
Now, Power Cycle Devices and confirm the action.





Waiting for Spanning Tree Protocol (STP)

Note: Because Packet Tracer also simulates the Spanning Tree Protocol (later), at times the switch may show amber lights on its interfaces. You will need to wait for the lights to turn green on the switches before they will forward any Ethernet frames.

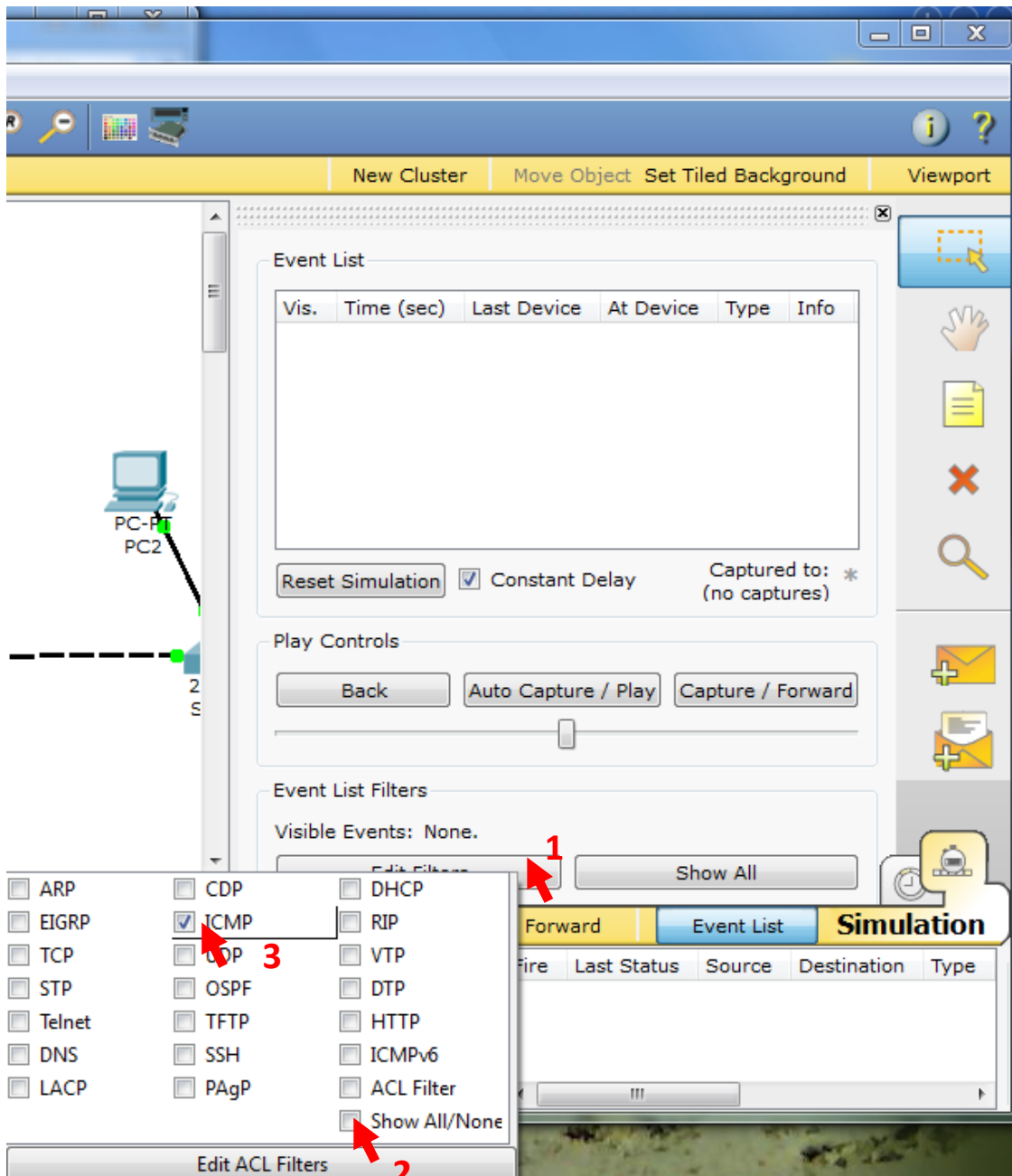


Step 8: Verifying Connectivity in Simulation Mode

Be sure you are in **Simulation** mode.



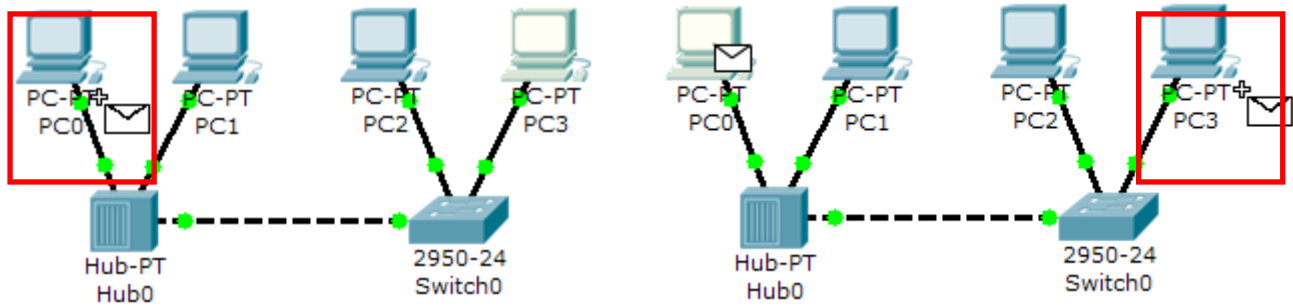
Deselect all filters (All/None) and select only **ICMP**.



Select the **Add Simple PDU** tool used to ping devices.



Click once on PC0, then once on PC3.



Continue clicking **Capture/Forward** button until the ICMP ping is completed. You should see the ICMP messages move between the hosts, hub and switch. The PDU **Last Status** should show as **Successful**. Click on **Clear Event List** if you do not want to look at the events or click **Preview Previous Events** if you do. For this exercise it does not matter.

Packet Tracer 5.0 by Cisco Systems, Inc.

Logical [Root] New Cluster Move Object Set Tiled Background Viewport

Event List

Vis.	Time (sec)	Last Device	At Device	Type	Info
	0.009	Switch0	PC3	ICMP	
	0.010	PC3	Switch0	ICMP	
	0.011	Switch0	Hub0	ICMP	
			PC0	ICMP	
			PC1	ICMP	

Buffer Full -- Packet Tracer 5.0

The maximum number of events has been reached. You may clear the event list and continue from where you left off or adjust the filters to view previous events.

Clear Event List View Previous Events

Event List Filters

Visible Events: ICMP

Edit Filters Show All

Time: 01:45:00.969 Power Cycle Devices PLAY Back Auto Capture / Play Capture / Forward Event List Simulation

Connections

Copper Cross-Over

Scenario 0

New Delete

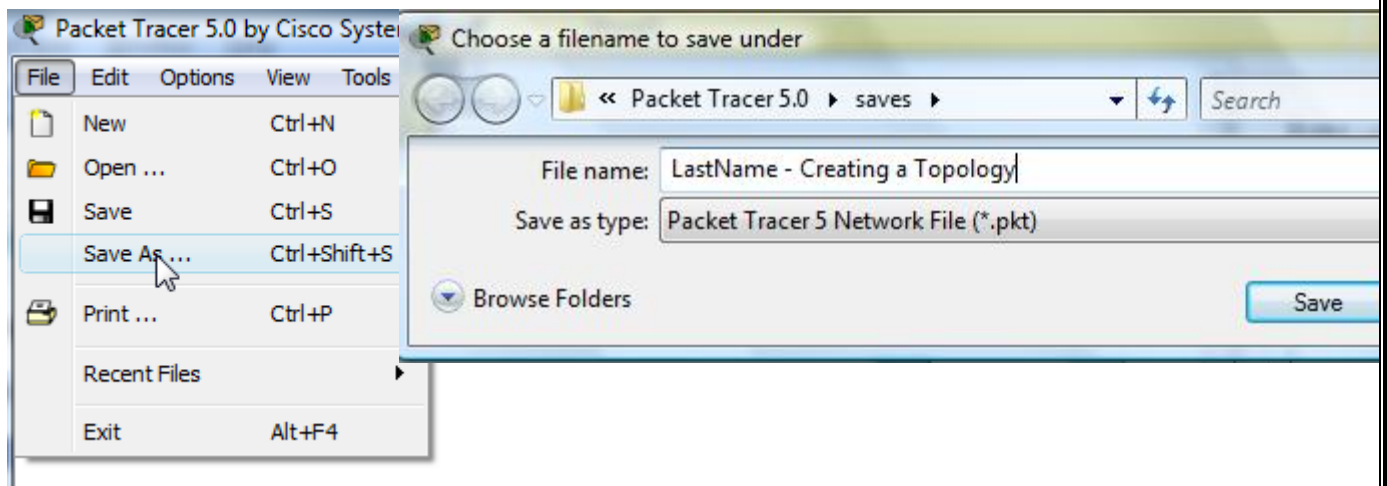
Toggle PDU List Window

Fire Last Status Source Destination Type

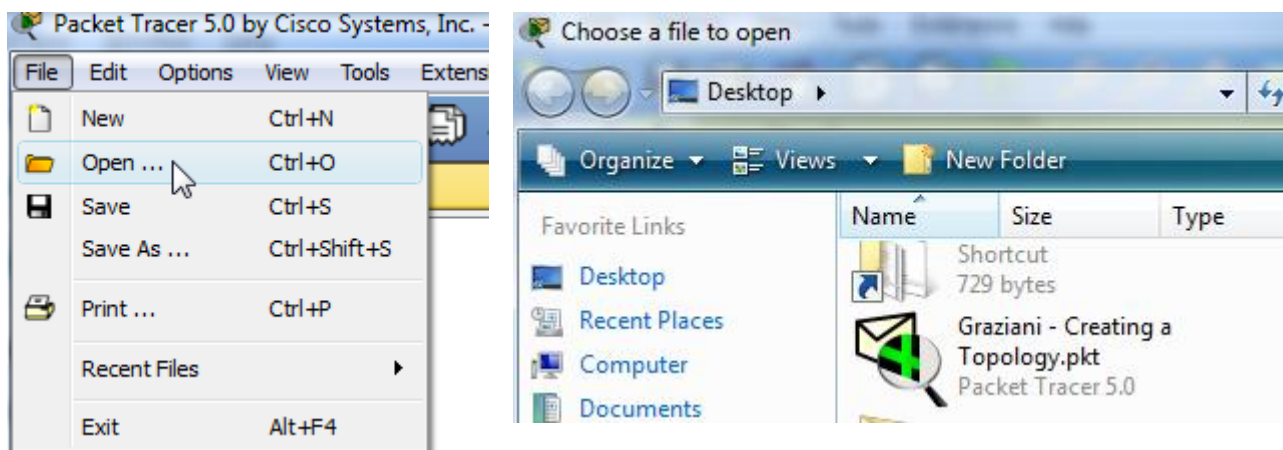
Successful PC0 PC3 ICMP

Step 9: Saving the Topology

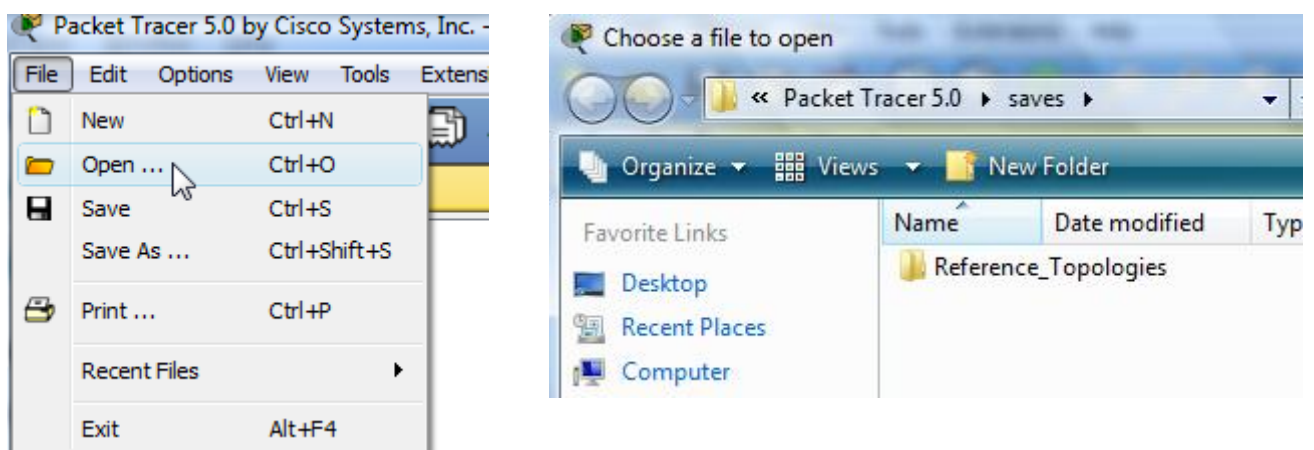
Perform the following steps to save the topology (uses .pkt file extension).

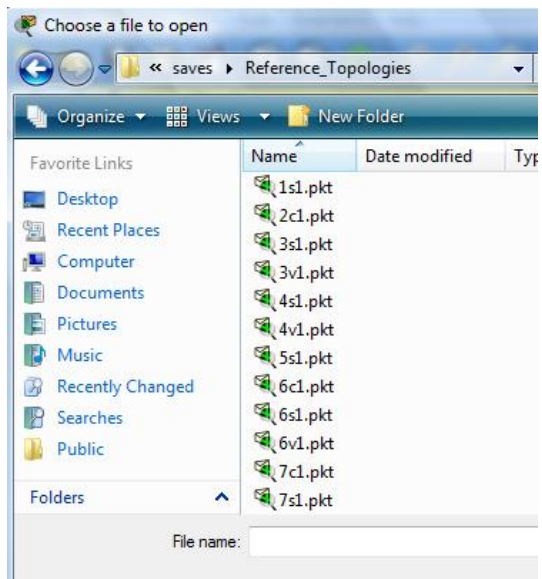


Opening Existing Topologies



Opening Existing PT Topologies



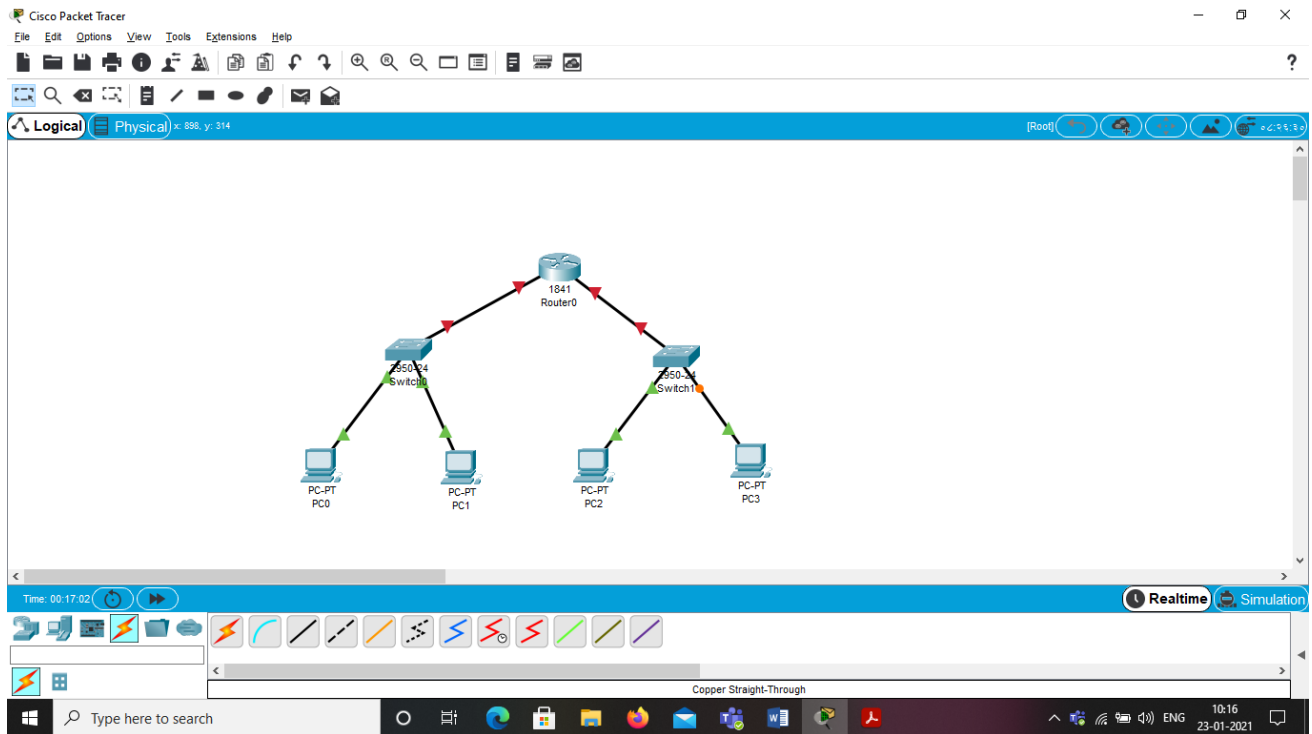


Experiment 3

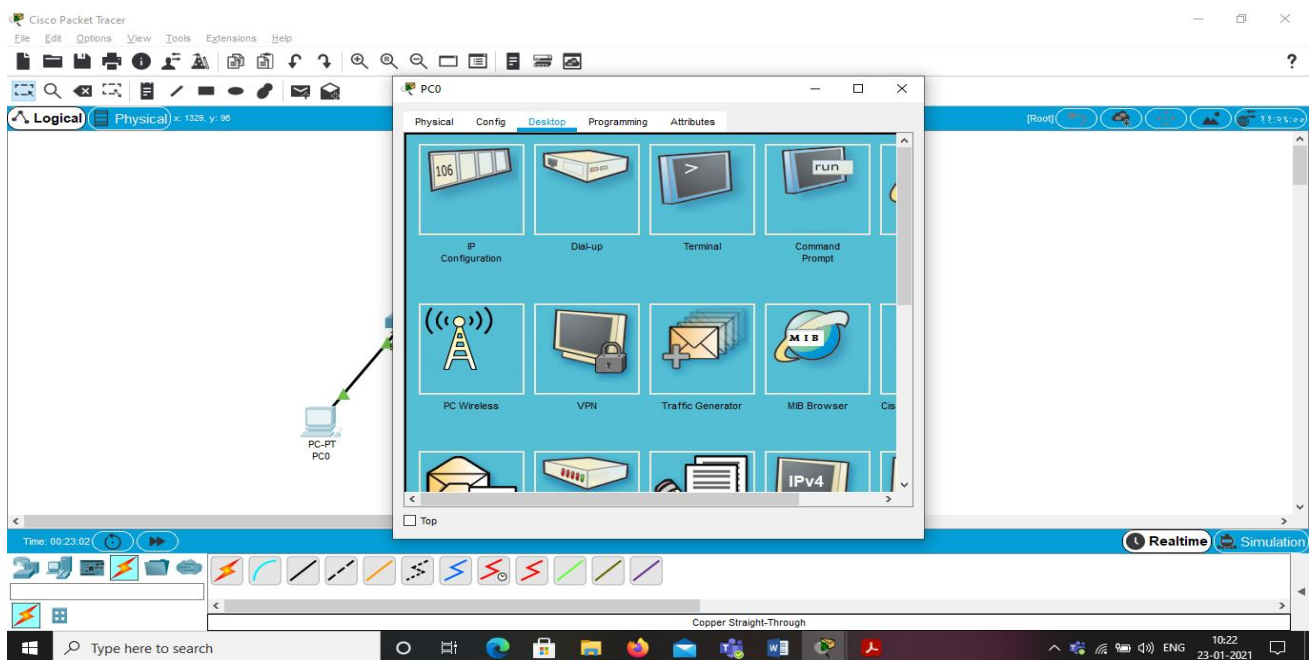
Basic Configuration of switch/router using Cisco packet tracer

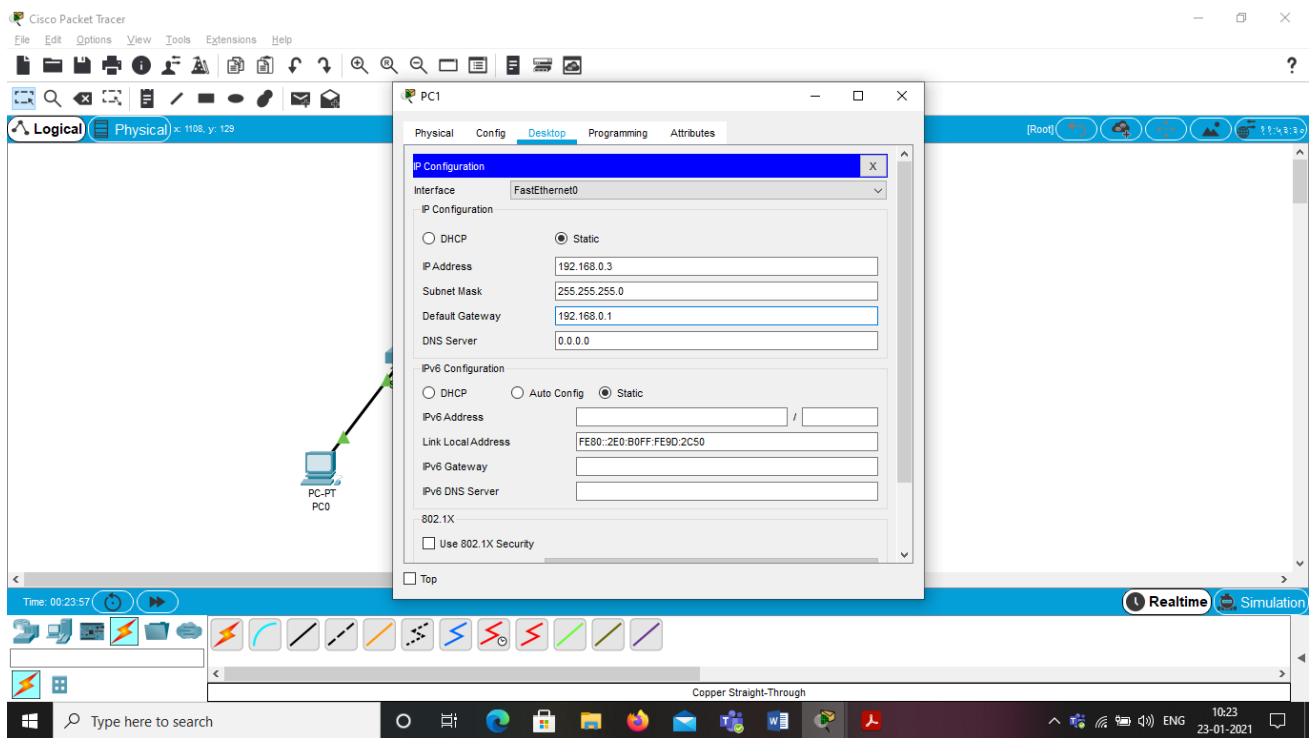
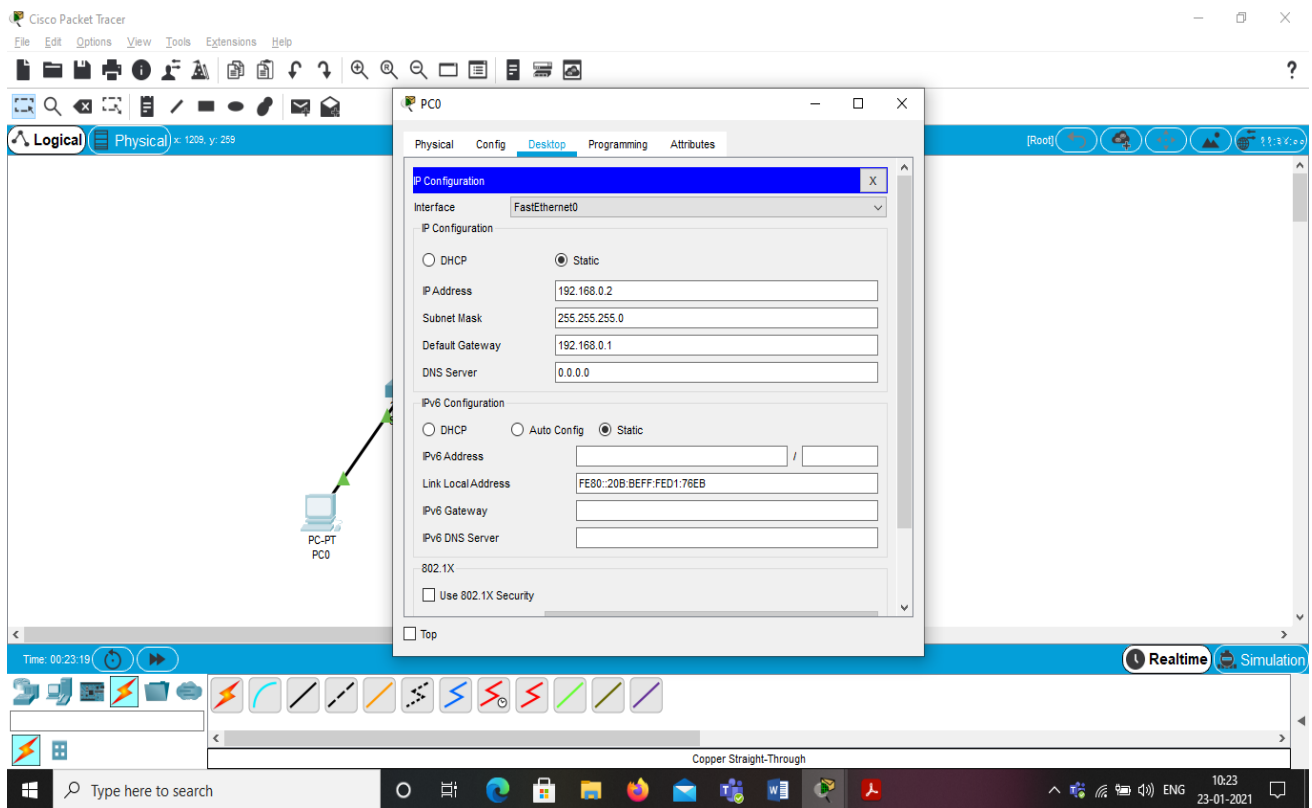
(a): Configuration of Router using cisco packet tracer

Step 1: Construct the topology

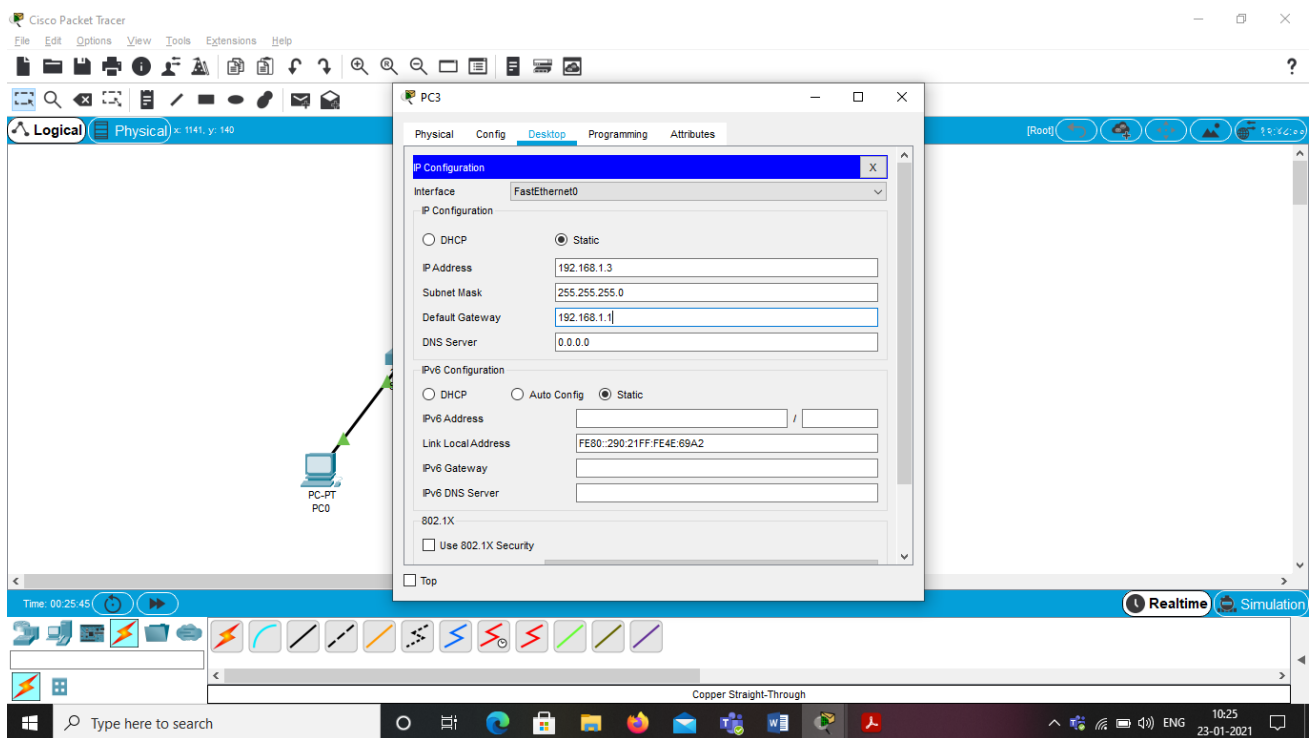
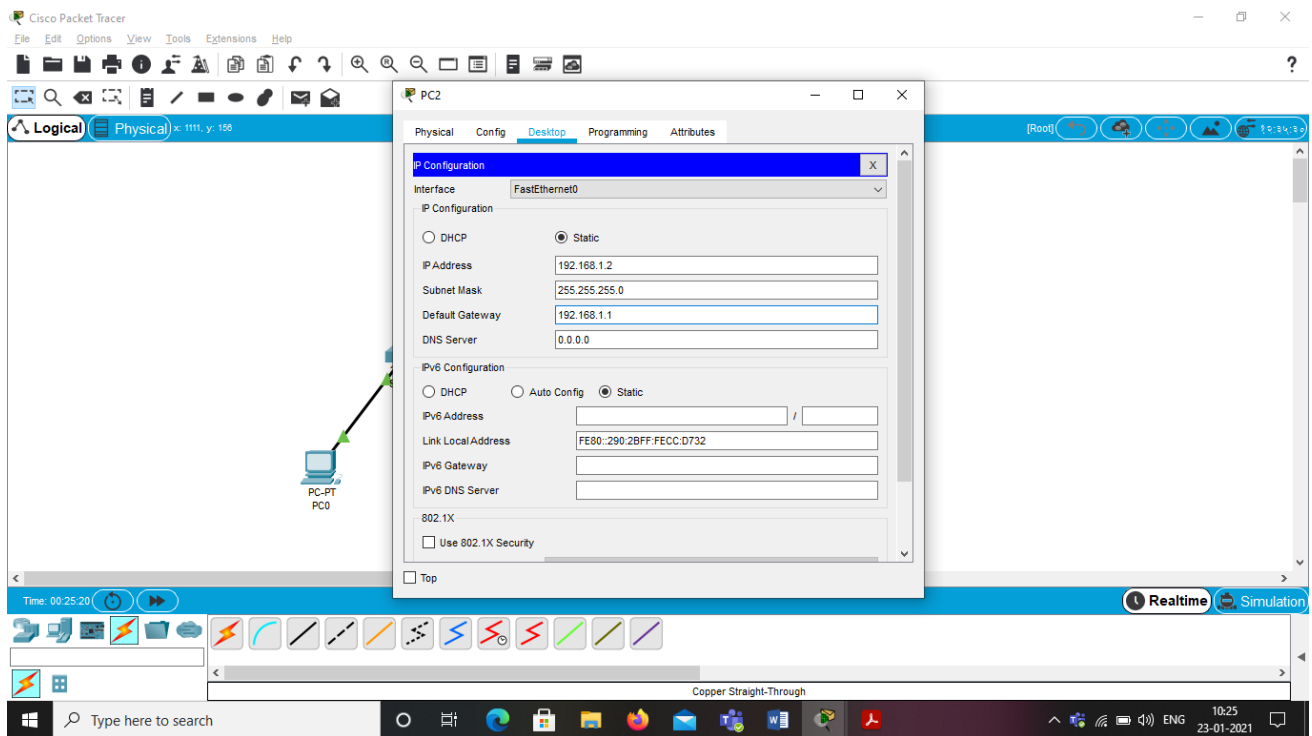


Step 2: Assign IP addresses to all PC's.



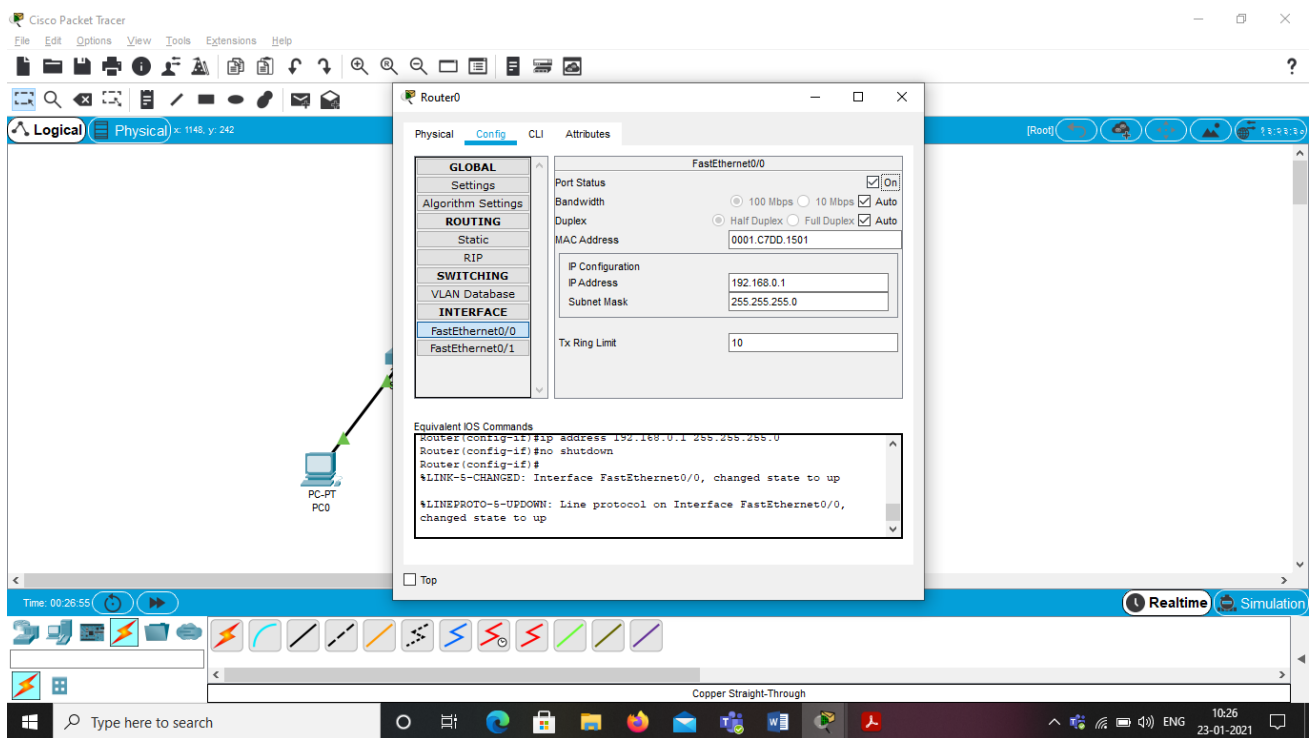


Use another network address for second network.

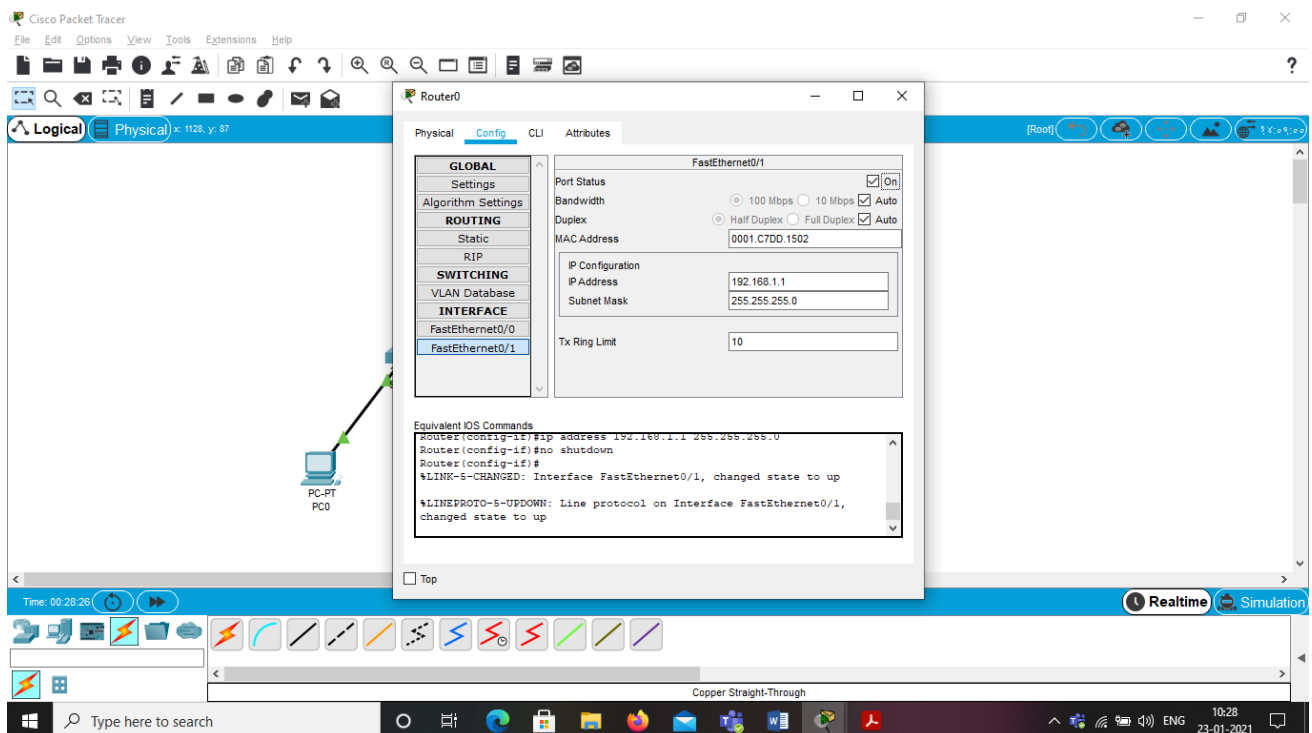


Step 3: Assign the IP address for router

Assign the gateway address of 1st network and don't forget to turn on the port status

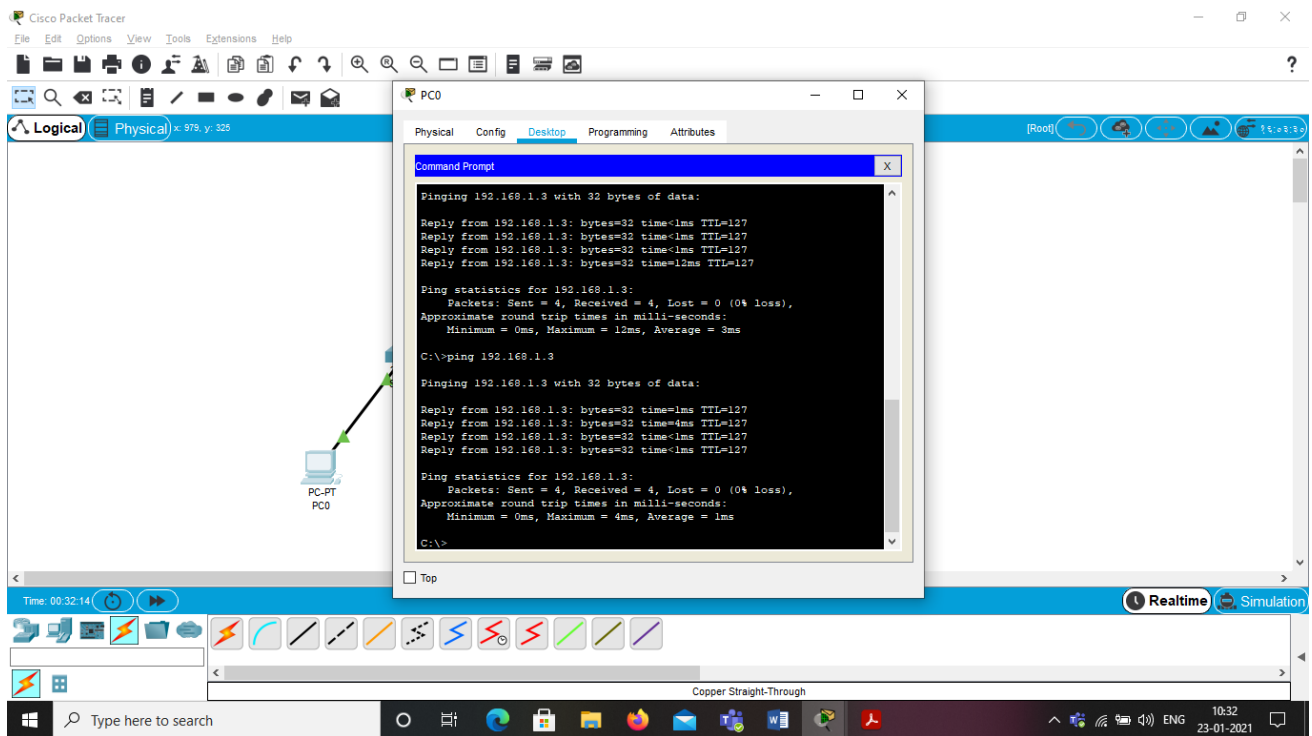


Assign the gateway address of 2nd network for FastEthernet0/1 interface

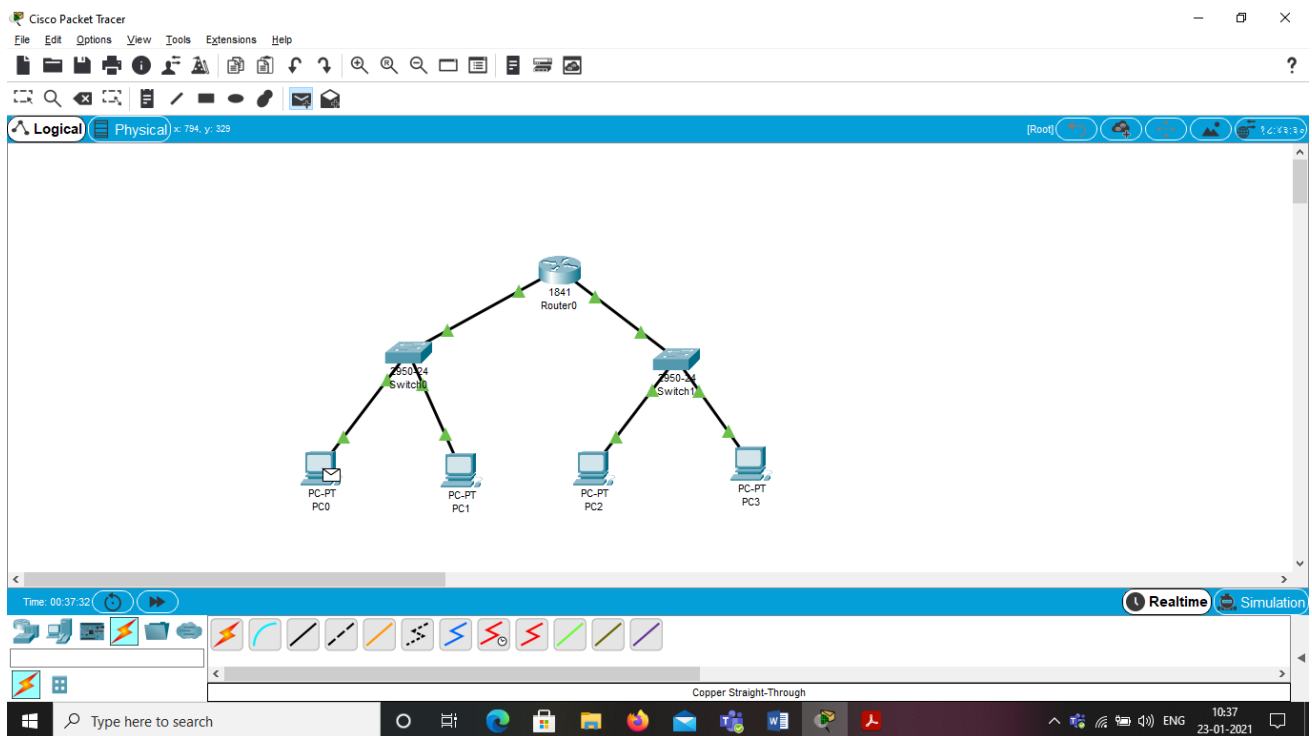


Step 3: Check the connectivity from one network to another network

Select any PC from 1st network go to Desktop tab->Command Prompt->execute ping command for the 2nd network.



Step 4: Send Simple PDU.



Step 5: Check in simulation mode

Cisco Packet Tracer

File Edit Options View Tools Extensions Help

Logical Physical x: 1025, y: 442

[Root]

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Router0	ICMP
	0.003	Router0	Switch1	ICMP

Reset Simulation ☒ Constant Delay Capturing...

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT-TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, RAGP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Time: 00:38:56.588 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Copper Straight-Through

Fire Last Status Source Destination Type Color Time(sec) Periodic Num

In Progress		PC0	PC3	ICMP		0.000	N	0
-------------	--	-----	-----	------	--	-------	---	---

10:39 23-01-2021

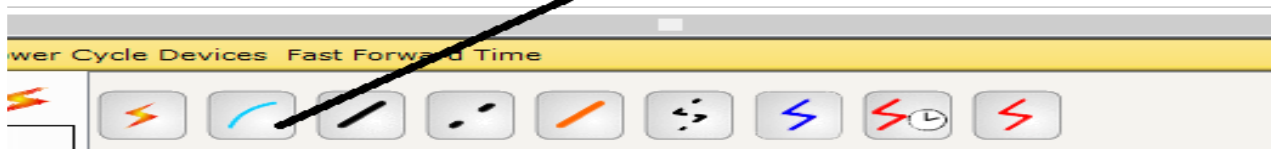
(b): Configuration of Switch using cisco packet tracer



Step 1:

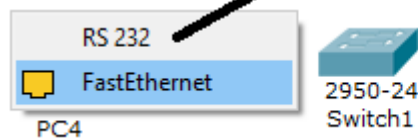


select console cable



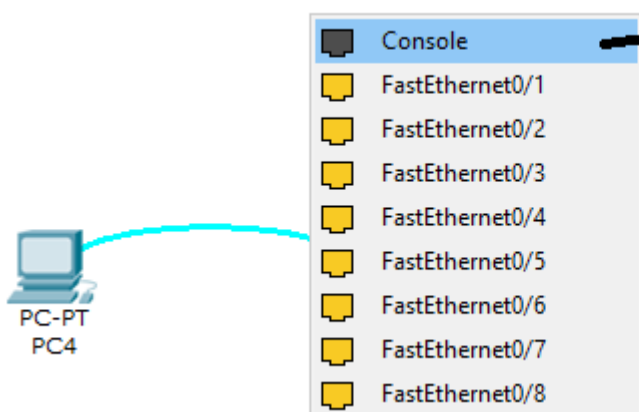
Step 2:

select RS232 in computer system



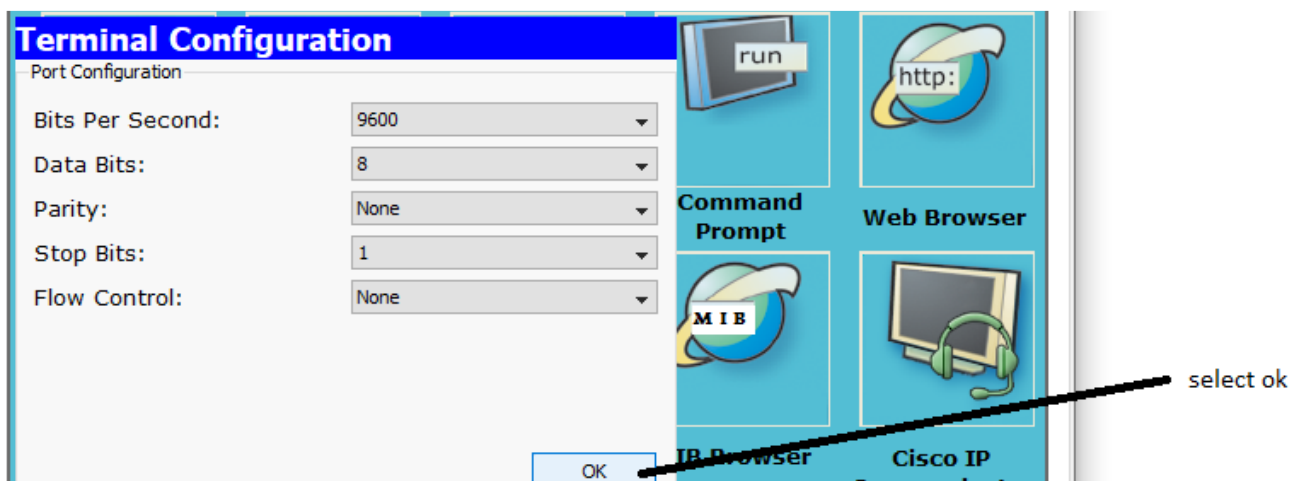
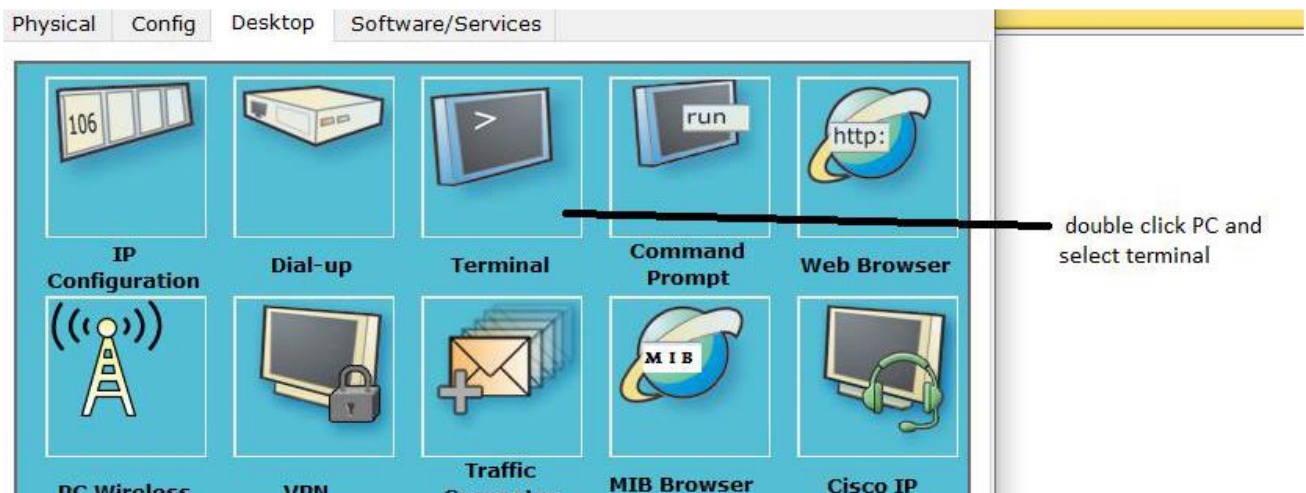
Step 3:

switch select console using console cable





Step 4:



Basic commands:

switch> ---> User Mode

switch>enable --> Enters into the Privilege mode

switch# --> Privilege mode

switch# configure terminal (or) conf t --> Enable Configuration Mode

switch(config)# --> Configuration Mode

Helping commands

switch> ? --> Help to list the available commands in this mode

switch>te? --> Lists all the commands starts with "tel"

switch# ? --> Help

Step 1: Erase the startup configuration file from NVRAM.

Type the erase startup-config command to remove the startup configuration from nonvolatile random access memory (NVRAM).

Switch # erase startup-config

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]

[OK]

Erase of nvram: complete

Router#

Step 2: Reload the switch.

Issue the reload command to remove an old configuration from memory. When prompted to Proceed with reload, press Enter to confirm the reload. Pressing any other key will abort the reload.

switch# reload

Proceed with reload? [confirm]

*Nov 29 18:28:09.923: %SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.

Note: You may receive a prompt to save the running configuration prior to reloading the router. Respond by typing no and press Enter.

System configuration has been modified. Save? [yes/no]: no

Step 3:

Use the show flash command to determine if any VLANs have been created on the switch.

Switch# show flash

Directory of flash:/

2 -rwx 1919 Mar 1 1993 00:06:33 +00:00 private-config.text

3 -rwx 1632 Mar 1 1993 00:06:33 +00:00 config.text

4 -rwx 13336 Mar 1 1993 00:06:33 +00:00 multiple-fs

5 -rwx 11607161 Mar 1 1993 02:37:06 +00:00 c2960-lanbasek9-mz.150-2.SE.bin

6 -rwx 616 Mar 1 1993 00:07:13 +00:00 vlan.dat

32514048 bytes total (20886528 bytes free)

Switch#

Step 4

Switch#

Switch> show version

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE, RELEASE SOFTWARE (fc1)

Technical Support: <http://www.cisco.com/techsupport>

Copyright (c) 1986-2012 by Cisco Systems, Inc.

Compiled Sat 28-Jul-12 00:29 by prod_rel_team

ROM: Bootstrap program is C2960 boot loader

BOOTLDR: C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(53r)SEY3, RELEASE SOFTWARE

(fc1)

Switch uptime is 2 minutes

System returned to ROM by power-on

System image file is "flash://c2960-lanbasek9-mz.150-2.SE.bin"

<output omitted>

Which IOS image version is currently in use by your switch?

Step 5: Configure the clock.

As you learn more about networking, you will see that configuring the correct time on a Cisco switch can be helpful when you are troubleshooting problems. The following steps manually configure the internal clock of the switch.

a. Display the current clock settings.

Switch> **show clock**

*00:30:05.261 UTC Mon Mar 1 1993

b. Configure the clock setting. The question mark (?) provides help and allows you to determine the expected input for configuring the current time, date, and year. Press Enter to complete the clock configuration.

Switch# **clock set ?**

hh:mm:ss Current Time

Switch# **clock set 15:08:00 ?**

<1-31> Day of the month

MONTH Month of the year

Switch# **clock set 15:08:00 Oct 26 ?**

<1993-2035> Year

Switch# **clock set 15:08:00 Oct 26 2012**

Switch#

*Oct 26 15:08:00.000: %SYS-6-CLOCKUPDATE: System clock has been updated from 00:31:43 UTC Mon Mar 1 1993 to 15:08:00 UTC Fri Oct 26 2012, configured from console by console.

c. Enter the show clock command to verify that the clock setting has updated.

Switch# **show clock**

15:08:07.205 UTC Fri Oct 26 2012

Step 6: Give the switch a name.

Use the hostname command to change the switch name to S1.

Switch(config)# hostname S1

S1(config)#

Step 7: Enter a login MOTD banner.

A login banner, known as the message of the day (MOTD) banner, should be configured to warn anyone accessing the switch that unauthorized access will not be tolerated. The banner motd command requires the use of delimiters to identify the content of the banner message. The delimiting character can be any character as long as it does not occur in the message. For this reason, symbols, such as the #, are often used.

S1(config)# banner motd #

Enter TEXT message. End with the character '#'

Unauthorized access is strictly prohibited and prosecuted to the full extent
of the law. #

S1(config)# exit

S1#

Step 8: Save the configuration.

Use the copy command to save the running configuration to the startup file on non-volatile random access memory (NVRAM).

S1# copy running-config startup-config

Destination filename [startup-config]? [Enter]

Building configuration...

[OK]

S1#

Step 9 : Display the current configuration.

The show running-config command displays the entire running configuration, one page at a time. Use the spacebar to advance paging.

S1# show running-config

Building configuration...

Current configuration : 1409 bytes

!

! Last configuration change at 03:49:17 UTC Mon Mar 1 1993

Step 10 : Display the status of the connected interfaces on the switch.

To check the status of the connected interfaces, use the show ip interface brief command. Press the spacebar to advance to the end of the list.

S1# show ip interface brief

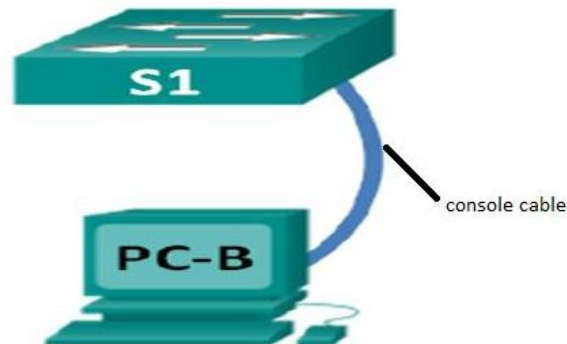
```
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset up up
FastEthernet0/1 unassigned YES unset up up
FastEthernet0/2 unassigned YES unset down down
FastEthernet0/3 unassigned YES unset down down
FastEthernet0/4 unassigned YES unset down down
FastEthernet0/5 unassigned YES unset down down
FastEthernet0/6 unassigned YES unset up up
FastEthernet0/7 unassigned YES unset down down
FastEthernet0/8 unassigned YES unset down down
FastEthernet0/9 unassigned YES unset down down
```

Step 11: Show vlan:

Show vlan command will display the VLANs. For administrative purpose, switch automatically create VLAN 1 and assign all its interfaces to it. You can create custom VLANs from global configuration mode and then assign them to interfaces.

Experiment 4

Configure the privilege level password and user authentication in switch.



I How to Set Hostname and Configure Console Password

1. To set the host name

```
Switch(config)# hostname CISCO
```

2. To set console password

```
CISCO( config)#
```

```
CISCO( config)#line console 0
```

```
CISCO( config-line)#password cisco123
```

```
CISCO( config-line)#login
```

```
CISCO( config-line)#exit
```

```
CISCO( config)#exit
```

```
CISCO(#exit
```

Check the Console Password

```
Press RETURN to get started!
```

```
User Access Verification
```

```
Password:
```

```
Password:
```

```
Switch>|
```

II How to Set Privilege level password

1)Set a privilege password

!!! Clear Text Password not encrypted(less priority)

CISCO(config)# enable password muscat

!!! Encrypted password (more Priority)

CISCO(config)# enable secret nizwa

2) Verify the privilege Password

CISCO(config)# exit

CISCO# exit

CISCO con0 is now available

Press RETURN to get started.

User Access Verification

!!! TYPE HERE LINE CONSOLE Password

Password:

CISCO>enable

!!! TYPE HERE Privilege Level Password

Password:

III How to Set User Authentication in Switch

1)Set user authentication

CISCO# conf t

CISCO(config)# line console 0

CISCO(config-line)# login local

CISCO(config-line)# exit

CISCO(config)#username network password pwd

2) Verify the Authentication

CISCO(config)# exit

CISCO# exit

User Access Verification

Username: network

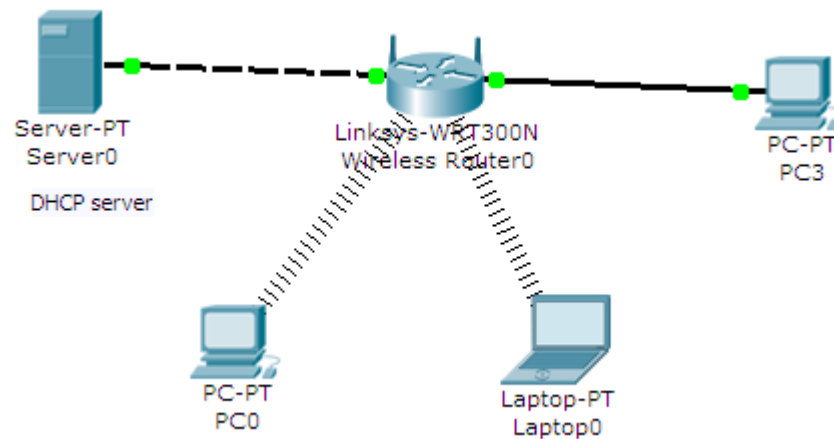
Password:

CISCO> enable

Password:

Experiment 5

Configure the DHCP Server and wireless router and check the connectivity



Step1 :configure the DHCP server

Server0

Physical Config Desktop Software/Services

GLOBAL

Settings

Algorithm Settings

SERVICES

HTTP

DHCP

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

INTERFACE

FastEthernet

DHCP

Service ☒ On ☐ Off

Pool Name

Default Gateway

DNS Server

Start IP Address :

Subnet Mask:

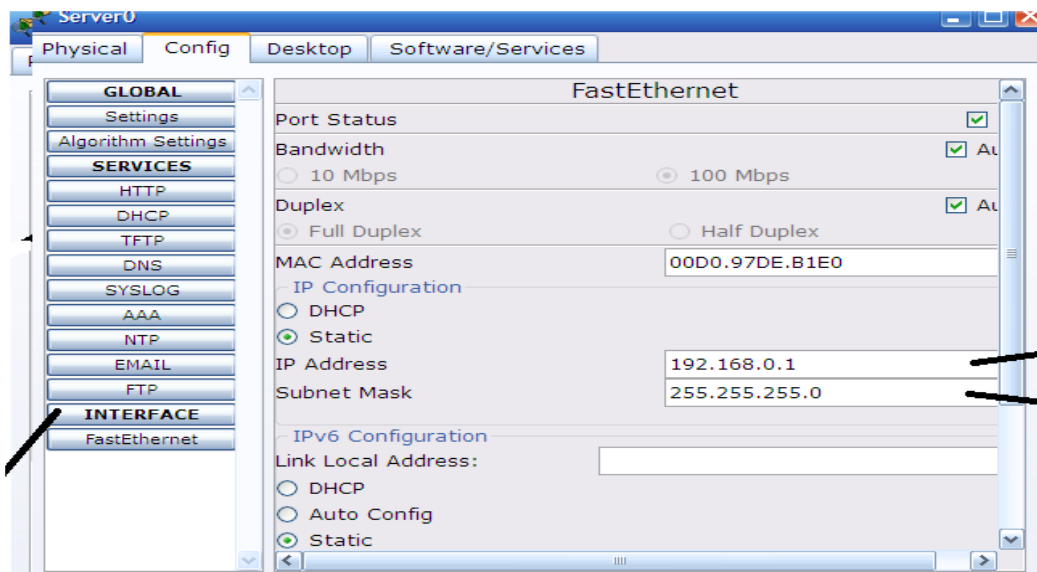
Maximum number of Users :

TFTP Server:

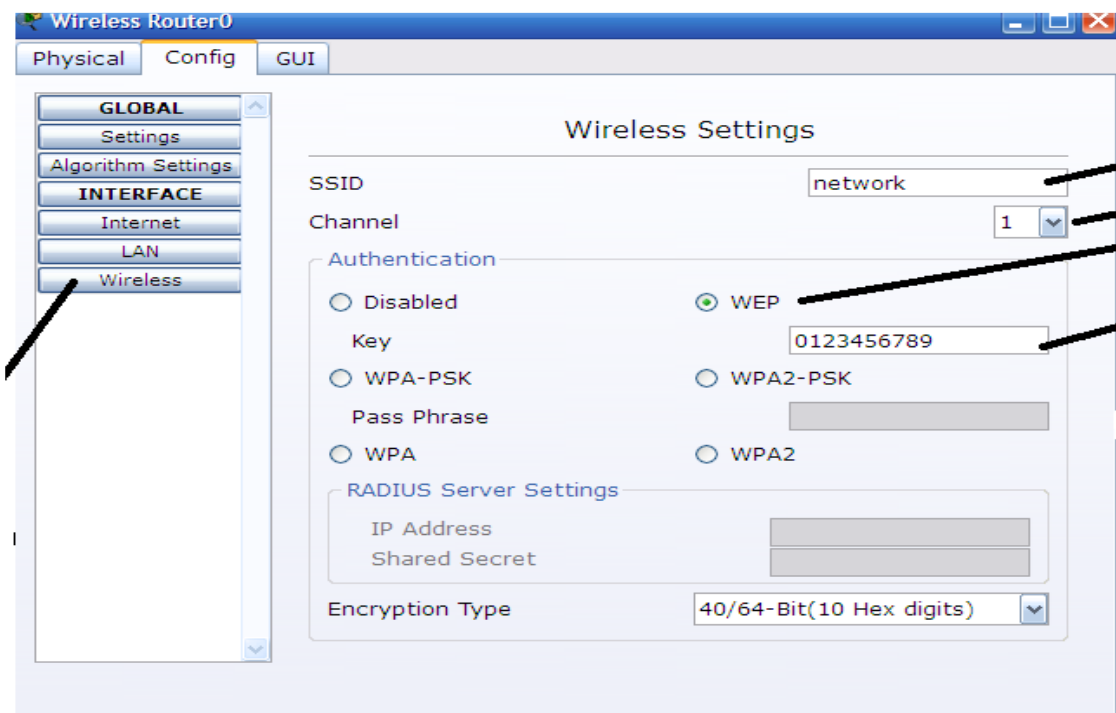
Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Maximum number of Users	TFTP Server
serv...	0.0.0.0	0.0.0.0	192.168...	255.2...	255	0.0.0.0

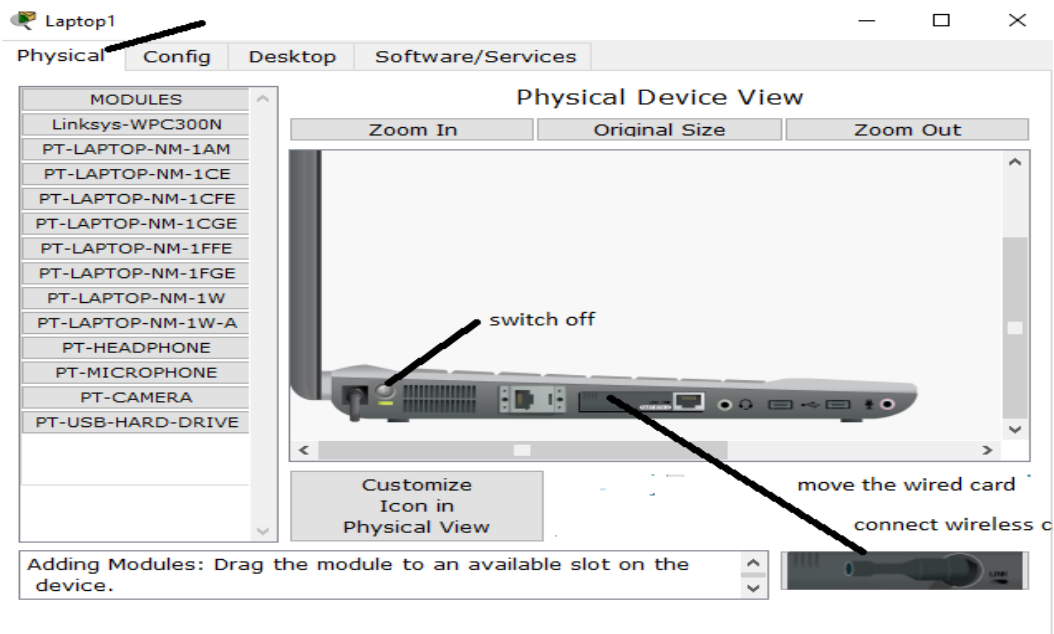
Step2: In server configure the fast Ethernet



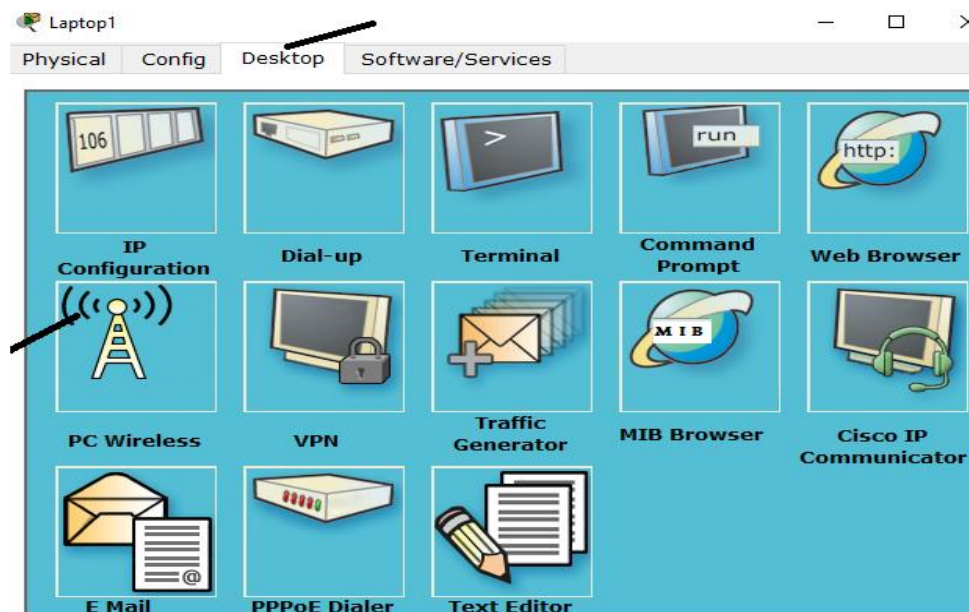
Step 3: configure the wireless router



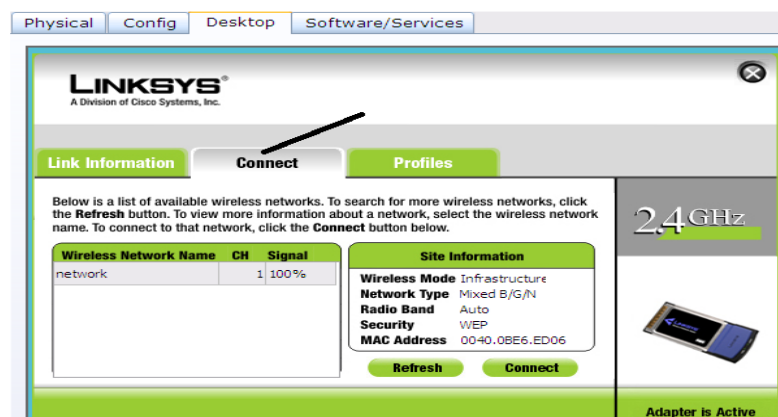
Step4 :Laptop remove the wired NIC card to wireless card



Step 5: Laptop connect the wireless



Step 6 : connect Laptop with wireless router



Step 7 : enter web key

Physical Config Desktop Software/Services

LINKSYS
A Division of Cisco Systems, Inc.

WEP Key Needed for Connection

This wireless network has WEP encryption enabled. To connect to this network, select the level of WEP encryption. Enter the required passphrase or WEP key in the appropriate field below. Then click the **Connect**.

Security WEP Please select the wireless security method used by your existing wireless network.

WEP 64-bit To use WEP encryption, select 64-bit or 128-bit

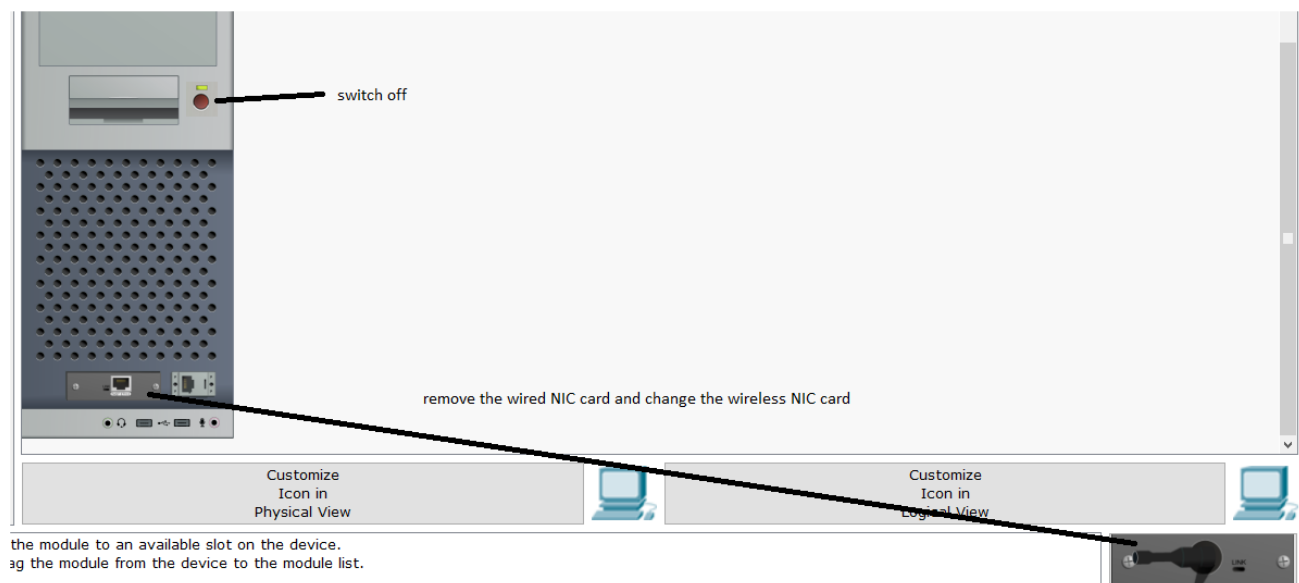
Passphrase The Passphrase is case-sensitive and should be no more than 16 characters in length.

WEP Key 1 012345678 When entering this manually, it should be 10 characters for 64-bit encryption or 26 characters for 128-bit encryption. Valid hexadecimal characters are "A" through "F" and numbers "0" through "9".

Cancel Connect

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC3001

Step8 : PC remove the wired NIC card to wireless card



Note : Same steps 5,6,7

Step 9: check the Pc and laptop for the DHCP ip address

IP Configuration

☒ DHCP
☐ Static

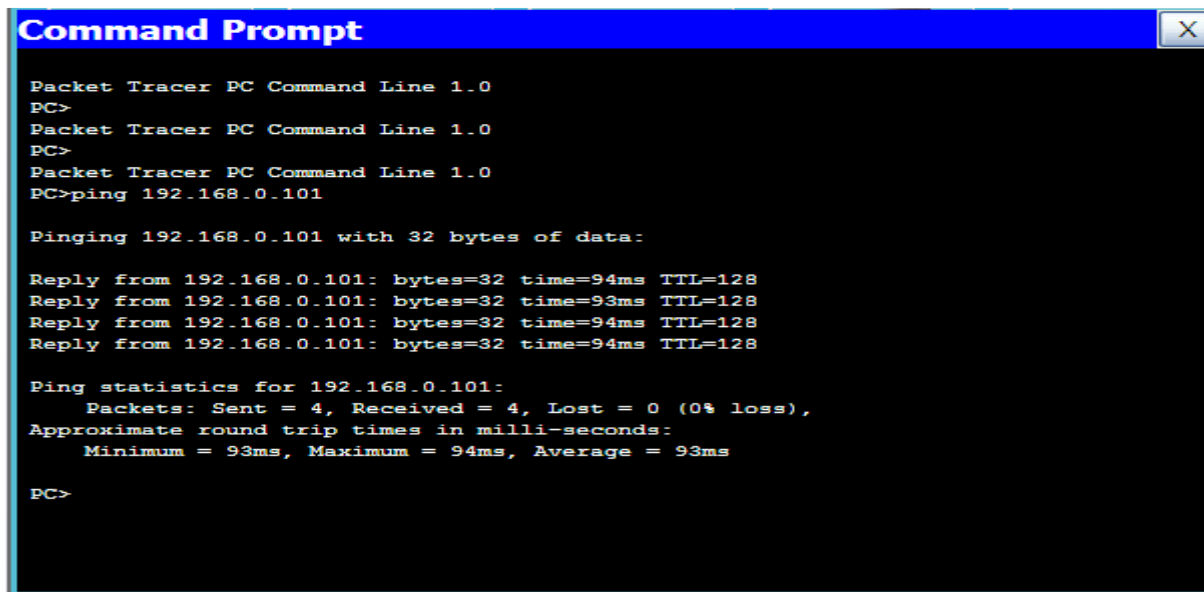
IP Address 192.168.0.102

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server

Step10: check the connectivity



```
Command Prompt
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.101

Pinging 192.168.0.101 with 32 bytes of data:

Reply from 192.168.0.101: bytes=32 time=94ms TTL=128
Reply from 192.168.0.101: bytes=32 time=93ms TTL=128
Reply from 192.168.0.101: bytes=32 time=94ms TTL=128
Reply from 192.168.0.101: bytes=32 time=94ms TTL=128

Ping statistics for 192.168.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 93ms, Maximum = 94ms, Average = 93ms

PC>
```